O-RAN.WG3.TS.E2SM-KPM-R004-v07.00

Technical Specification

O-RAN Work Group 3
Near-Real-time RAN Intelligent Controller
E2 Service Model (E2SM)
KPM

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Contents

Foreword

This Technical Specification (TS) has been produced by WG3 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

version xx.yy.zz

where:

xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.

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zz: the third digit-group included only in working versions of the document indicating incremental changes during the editing process. External versions never include the third digit-group. Always 2 digits with leading zero if needed.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies the E2 Service Model (E2SM) “Key Performance Measurement” (KPM) for the RAN function handling reporting of the cell-level performance measurements for 5G networks defined in TS 28.552 [4] and for EPC networks defined in TS 32.425 [8], and their possible adaptation of UE-level or QoS flow-level measurements.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

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The following referenced documents are necessary for the application of the present document.

[1] Void

[2] O-RAN-WG3.TS.E2GAP: “O-RAN E2 General Aspects and Principles”.

[3] O-RAN.WG3.TS.E2AP, "O-RAN E2 Application Protocol (E2AP) "

[4] 3GPP TS 28.552: "Management and orchestration 5G performance measurements".

[5] ITU-T Recommendation X.680 (2002-07): "Information technology – Abstract Syntax Notation One (ASN.1): Specification of basic notation".

[6] ITU-T Recommendation X.681 (2002-07): "Information technology – Abstract Syntax Notation One (ASN.1): Information object specification".

[7] ITU-T Recommendation X.691 (2002-07): "Information technology - ASN.1 encoding rules - Specification of Packed Encoding Rules (PER)".

[8] 3GPP TS 32.425: "Telecommunication management Performance Management Performance managements".

[9] IETF RFC 5905 (2010-06): "Network Time Protocol Version 4: Protocol and Algorithms Specification".

[10] 3GPP TS 32.404: "Telecommunication management; Performance Management (PM); Performance measurements; Definitions and template".

[11] Void

[12] O-RAN-WG3.TS.E2SM: "O-RAN E2 Service Model (E2SM)".

[13] 3GPP TS 36.413: "E-UTRAN; S1 Application Protocol (S1AP)"

[14] 3GPP TS 23.501: "System architecture for the 5G System (5GS); Stage 2"

[15] 3GPP TS 36.214: "E-UTRA; Physical layer; Measurements"

[16] 3GPP TS 38.215: "NR; Physical layer measurements"

[17] 3GPP TS 38.133: "NR; Requirements for support of radio resource management"

[18] 3GPP TS 38.214: "NR; Physical layer procedures for data"

[19] 3GPP TS 23.203: "Policy and charging control architecture"

[20] 3GPP TS 23.003: "Numbering, addressing and identification"

[21] O-RAN.WG3.TS.UCR: "O-RAN Use Cases and Requirements"

[22] Void

[23] O-RAN.WG1.OAD: "O-RAN Architecture Description"

[24] Void

[25] 3GPP TS 37.340: "NR; Multi-connectivity; Overall description; Stage-2".

2.2 Informative references

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References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] O-RAN.WG1.mMIMO: "O-RAN Massive MIMO Use Cases Technical Report"

[i.2] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications"

[i.3] 3GPP TR 25.921: "Guidelines and principles for protocol description and error handling".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms and definitions given in TR 21.905 [i.2], O-RAN WG1.OAD [23], O-RAN.WG3.TS.E2SM [12] and the following apply:

KPM Report: Key performance measurements for 4G LTE and 5G NR Network Functions.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 **Error! Reference source not found.**, O-RAN WG1.OAD [23] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 **Error! Reference source not found.** and O-RAN WG1.OAD [23].

EN-DC E-UTRA-NR Dual Connectivity
MR-DC Multi-Radio Dual Connectivity

4 General

4.1 Forwards and Backwards Compatibility

The forwards and backwards compatibility of the protocol is assured by a mechanism where all current and future messages, and IEs or groups of related IEs, include ID and criticality fields that are coded in a standard format that will not be changed in the future. These parts can always be decoded regardless of the standard version.

4.2 Specification Notations

For the purposes of the present document, the following notations apply:

Service When referring to a Service in the present document the **SERVICE NAME** is written with upper case characters and in bold followed by the word "service", e.g. **REPORT** service.

Procedure When referring to an elementary procedure in the present document the Procedure Name is written with the first letters in each word in upper case characters followed by the word "procedure", e.g. Handover Preparation procedure.

Message When referring to a message in the present document the MESSAGE NAME is written with all letters in upper case characters followed by the word "message", e.g. HANDOVER REQUEST message.

IE When referring to an information element (IE) in the present document the *Information Element Name* is written with the first letters in each word in upper case characters and all letters in Italic font followed by the abbreviation "IE", e.g. *E-RAB ID* IE.

Value of an IE When referring to the value of an information element (IE) the present document the "Value" is written as it is specified in the present document enclosed by quotation marks, e.g. "Value".

4.3 Void

5 E2SM Services

As defined in E2 General Aspects and Principles [2], a given RAN Function offers a set of services to be exposed over the E2 (**REPORT**, **INSERT**, **CONTROL, POLICY** and/or **QUERY**) using E2AP [3] defined procedures. Each of the E2AP Procedures listed in table 5-1 contains specific E2 Node RAN Function dependent Information Elements (IEs).

Table 5-1: Relationship between E2SM services and E2AP Information elements

RAN Function specific E2AP Information Elements	E2AP Information Element reference	Related E2AP Procedures
RIC Event Trigger Definition IE	E2AP [3] clause 9.2.9	RIC Subscription
RIC Action Definition IE	E2AP [3] clause 9.2.12	RIC Subscription
RIC Indication Header IE	E2AP [3] clause 9.2.17	RIC Indication
RIC Indication Message IE	E2AP [3] clause 9.2.16	RIC Indication
RIC Call Process ID IE	E2AP [3] clause 9.2.18	RIC Indication RIC Control
RIC Control Header IE	E2AP [3] clause 9.2.20	RIC Control
RIC Control Message IE	E2AP [3] clause 9.2.19	RIC Control
RIC Control Outcome IE	E2AP [3] clause 9.2.25	RIC Control
RAN Function Definition IE	E2AP [3] clause 9.2.23	E2 Setup RIC Service Update
RIC Query Header IE	E2AP [3] clause 9.2.36	RIC Query
RIC Query Definition IE	E2AP [3] clause 9.2.37	RIC Query
RIC Query Outcome IE	E2AP [3] clause 9.2.38	RIC Query

All of these RAN Function specific IEs are defined in E2AP [3] as “OCTET STRING”.

The purpose of this specification is to define the contents of these fields for the specific RAN Function “KPM Monitor”.

6 RAN Function Service Model Description

6.1 RAN Function Overview

E2 Service Model KPM (E2SM-KPM) shall support O-CU-CP, O-CU-UP, and O-DU as part of NG-RAN connected to 5GC or as part of E-UTRAN connected to EPC.

The E2 Node shall host the RAN Function "KPM Monitor" which performs the following functionalities:

- Exposure of available measurements from O-DU, O-CU-CP, and/or O-CU-UP via the RAN Function Definition IE.
- Periodic reporting of measurements subscribed from Near-RT RIC.

This E2SM specification also exposes a set of services described in clause 6.2.

6.2 Supported RIC Services

6.2.1 REPORT

The "KPM Monitor" RAN Function shall provide the following **REPORT** services:

a) Fundamental level services:

- E2 Node Measurement
- E2 Node Measurement for a single UE
- Condition-based, UE-level E2 Node Measurement
- Common Condition-based, UE-level E2 Node Measurement
- E2 Node Measurements for multiple UEs

b) Integrated level services:

- Multi-report

These services shall be initiated according to:

- Periodical event.

7 RAN Function Description

7.1 Description

The E2AP [3] procedures, E2 SETUP and RIC SERVICE UPDATE, are used to transport the RAN Function Definition IE.

For a specific RAN Function declared using E2SM-KPM, the *RAN Function Definition* IE, defined in clause 8.2.2, shall report the following information:

- RAN Function name along with associated information on E2SM definition
- Event trigger styles list along with the corresponding encoding type for each associated E2AP IE.
- RIC REPORT Service styles list along with the corresponding encoding type for each associated E2AP IE.

For the case where *RAN Function Definition* IE, defined in clause 8.2.2, is present in the E2 SETUP REQUEST message the IE shall provide a complete list of all Styles and associated Formats and, where appropriate, Measurement information for all supported RIC services reflecting the current status of the RAN Function.

For the case where *RAN Function Definition* IE, defined in clause 8.2.2, is present in the RIC SERVICE UPDATE message within the E2AP *RAN Functions Added List* IE, the IE shall provide a complete list of all Styles and associated Formats and, where appropriate, Measurement information for all supported RIC services for the newly added RAN Function with a new RAN Function ID.

For the case where *RAN Function Definition* IE, defined in clause 8.2.2, is present in the RIC SERVICE UPDATE message within the E2AP *RAN Functions Modified List* IE, the IE shall provide a complete list of all the Styles and associated Formats and, where appropriate, Measurement information for all supported RIC services including both modified and unchanged information for an existing RAN Function.

7.2 RAN Function Name

RAN Function Short Name "ORAN-E2SM-KPM"

RAN Function Description "KPM Monitor"

RAN Function Instance, required when and if an E2 Node exposes more than one instance of a RAN Function based on this E2SM.

7.3 Supported RIC Event Trigger Styles

7.3.1 Event Trigger Style Types

RIC Style Type	Style Name	Supported RIC Service Style			Style Description
		Report	Insert	Policy	
1	Periodic Report	1-5, 255	-	-	*RIC Event Trigger Definition* IE based on reporting period
2	Measurement Event	1-5	-	-	*RIC Event Trigger Definition* IE based on triggering condition(s) associated with specific measurements subscribed.

7.3.2 Event Trigger Style 1: Periodic Report

Event Trigger style 1 shall set the KPM report period and uses the *RIC Event Trigger Definition* IE Format 1 (see clause 8.2.1.1.1)

7.3.3 Event Trigger Style 2: Measurement Event

Event Trigger style 2 is used to configure event-based reporting based on measurement data. Each event condition is defined using measurement-based or value-based parameters, and is evaluated based on the availability of measurement samples. If the *Event Evaluation Period* IE is included, the event condition is evaluated periodically according to the configured evaluation period. Otherwise, it is continuously evaluated whenever a new measurement sample becomes available for the associated parameters. The E2 Node can be configured with multiple event conditions simultaneously and triggered to send KPM reports when all the configured events happen or for any logical combination of events.

Example For a window-average of three measurement samples of a specific measurement associated with a cell to be compared against a threshold integer value and triggered when exceeding the threshold, the *Event Condition* IE in 8.2.1.1.2 can be defined as follows:

- The *Event Left Expression* IE contains only one parameter and selects the *Parameter Type* IE as "Measurement", where the *Measurement Definition* IE specifies the measurement and the associated cell via 8.3.33, and the corresponding *Window Definition* IE configures a window size of three with its aggregation function set to "avg".
- The *Event Logical Operator* IE is set to "greaterthan".
- The *Event Right Expression* IE contains only one parameter and selects the *Parameter Type* IE as "Value", where the *Test Value* IE is assigned to the threshold integer value.

Example For a sum of two different measurements (each from measurement type 1 and type 2, respectively) associated with a cell to be compared against a window-average of five measurement samples of a measurement type 3 associated with the same cell, and triggered when exceeding the window-averaged value, the *Event Condition* IE in 8.2.1.1.2 can be defined as follows:

- The *Event Left Expression* IE contains two parameters and selects the first *Parameter Type* IE as "Measurement", where the *Measurement Definition* IE specifies the measurement type 1 and the associated cell via 8.3.33 while the corresponding *Window Definition* IE is absent.
- The *Event Left Expression* IE also selects the second *Parameter Type* IE as "Measurement", where the *Measurement Definition* IE specifies the measurement type 2 and the associated cell via 8.3.33 while the corresponding *Window Definition* IE is absent. The *Event Left Expression* IE also includes the *Aggregation Function* IE set to "sum".
- The *Event Logical Operator* IE is set to "greaterthan".
- The *Event Right Expression* IE contains only one parameter and selects the *Parameter Type* IE as "Measurement", where the *Measurement Definition* IE specifies the measurement type 3 and the associated cell via 8.3.33, and the corresponding *Window Definition* IE configures a window size of five with its aggregation function set to "avg".

For any measurement-based parameter used in an event condition defined by the *RIC Event Trigger Definition* IE, the corresponding measurement must be subscribed by the *RIC Action Definition* IE in the same RIC SUBSCRIPTION REQUEST message from the Near-RT RIC.

In this version of the specification, a measurement-based parameter associated with a specific UE is not supported.

This *RIC Event Trigger Definition* IE style uses *RIC Event Trigger Definition* IE Format 2 (8.2.1.1.2).

7.4 Supported RIC REPORT Service Styles

7.4.1 REPORT Service Style Type

RIC Style Type	Style Name	Style Type Description
1	E2 Node Measurement	Used to carry measurement report from a target E2 Node Belongs to Fundamental level REPORT Services
2	E2 Node Measurement for a single UE	Used to carry measurement report for a single UE of interest from a target E2 Node Belongs to Fundamental level REPORT Services
3	Condition-based, UE-level E2 Node Measurement	Used to carry UE-level measurement report for a group of UEs per measurement type matching subscribed conditions from a target E2 Node Belongs to Fundamental level REPORT Services
4	Common Condition-based, UE-level Measurement	Used to carry measurement report for a group of UEs across a set of measurement types satisfying common subscribed conditions from a target E2 Node Belongs to Fundamental level REPORT Services
5	E2 Node Measurement for multiple UEs	Used to carry measurement report for multiple UEs of interest from a target E2 Node Belongs to Fundamental level REPORT Services
255	Multiple report measurements	Used for multiple actions of the selected fundamental level REPORT Service style(s). Belongs to Integrated level REPORT Services.

The Report Service style 255 supports integrated level services configured per *RIC Action Definition* message by reusing the report *RIC Action Definition* IE and *RIC Indication Message* IE messages defined in the selected fundamental level Report Service style(s).

7.4.2 REPORT Service Style 1: E2 Node Measurement

7.4.2.1 REPORT Service Style description

REPORT Service style 1 provides the performance measurement information collection from an E2 Node.

7.4.2.2 REPORT Service *RIC Action Definition* IE contents

REPORT Service style 1 aims to subscribe to the measurements defined in TS 28.552 [4] and TS 32.425 [8], and shall use the *RIC Action Definition* IE Format 1 (see clause 8.2.1.2.1).

REPORT Service *RIC Action Definition* IE Format 1 contains measurement types that Near-RT RIC is requesting to subscribe followed by a list of subcounters to be measured for each measurement type, and a granularity period indicating collection interval of those measurements. For a certain measurement type to be subscribed, the *Matching Condition for Reporting* IE may also be included to give conditions for the E2 Node in reporting measurements, to report a measured value corresponding to that measurement type only when the value satisfies the given conditions.

For the measurement types that belong to a measurement object class confined in a single cell (e.g. "EUtranCellFDD" in TS 32.425 [8] or "NRCellDU" in TS 28.552 [4]), the *Cell Global ID* IE shall be included in the IE to point to a specific cell for collecting measurements within the E2 Node. The *Cell Global ID* IE shall not be included if all the subscribed measurement types are cell agnostic, i.e. belonging to measurement object classes not confined in a single cell (e.g. "GNBCUUPFunction" in TS 28.552 [4]). In case that both single-cell-confined and cell agnostic measurement types are subscribed together, the *Cell Global ID* IE shall be included in the IE and the E2 Node shall ignore the included *Cell Global ID* IE for those cell agnostic measurement types.

A measurement ID can be used for subscription instead of a measurement type if an identifier of a certain measurement type was exposed by an E2 Node via the *RAN Function Definition* IE.

7.4.2.3 REPORT Service *RIC Indication Header* IE contents

REPORT Service style 1 shall use the *RIC Indication Header* IE Format 1 (see clause 8.2.1.3.1), which contains a measurement collection start time as UTC format.

REPORT Service *RIC Indication Header* IE Format 1 may carry file format version, sender name, sender type, and vendor name as printable strings.

7.4.2.4 REPORT Service *RIC Indication Message* IE contents

REPORT Service style 1 shall use the *RIC Indication Message* IE Format 1 (see clause 8.2.1.4.1).

REPORT Service *RIC Indication Message* IE Format 1 carries a set of measurement data reported from an E2 Node. The reported data contains a set of measurement records, each collected at every granularity period during the reporting period. In case that the *Matching Condition for Reporting* IE has been configured in the *RIC Action Definition* IE for a measurement type and a measured value corresponding to that measurement type in a granularity period did not satisfy the configured reporting condition, the E2 Node shall indicate “Not Satisfied” instead of reporting the actual measured value within the measurement record corresponding to that granularity period. In case the configured reporting condition(s) are not satisfied entirely for a reporting period for all the subscribed measurement types, the E2 Node may omit sending the report for this reporting period. In case the E2 Node is not able to provide reliable data for a granularity period during the reporting period, it may include the optional *Incomplete Flag* IE, which indicates that the corresponding measurements record in the reported data is not reliable.

REPORT Service *RIC Indication Message* IE Format 1 may carry subscription information, i.e. *Measurement Information List* IE that indicates the order of measured values for each measurement record in the reported data, or their granularity period. If not present, the original subscription information shall apply.

7.4.3 REPORT Service Style 2: E2 Node Measurement for a single UE

7.4.3.1 REPORT Service Style description

REPORT Service style 2 provides the performance measurement information collection for a single UE of interest from an E2 Node.

7.4.3.2 REPORT Service *RIC Action Definition* IE contents

REPORT Service style 2 shall use the *RIC Action Definition* IE Format 2 (see clause 8.2.1.2.2), where the included UE ID indicates a specific UE of interest for measurement collection.

The rest of the subscription information follows the same as described in 7.4.2.2.

7.4.3.3 REPORT Service *RIC Indication Header* IE contents

REPORT Service style 2 shall use the *RIC Indication Header* IE Format 1 (see clause 8.2.1.3.1) as described in 7.4.2.3.

7.4.3.4 REPORT Service *RIC Indication Message* IE contents

REPORT Service style 2 shall use the *RIC Indication Message* IE Format 1 (see clause 8.2.1.4.1) as described in 7.4.2.4, where the measurement data reported is associated only to a specific UE that was subscribed.

7.4.4 REPORT Service Style 3: Condition-based, UE-level E2 Node Measurement

7.4.4.1 REPORT Service Style description

REPORT Service style 3 provides the UE-level performance measurement information collection for a group of UEs per measurement type matching subscribed conditions from an E2 Node.

7.4.4.2 REPORT Service *RIC Action Definition* IE contents

REPORT Service style 3 shall use the *RIC Action Definition* IE Format 3 (see clause 8.2.1.2.3), where, for each requested measurement within the *Measurement Information Condition List* IE, the *Matching Condition* IE serves as a condition to include the matched UEs' measurement values into the reporting. The *Matching Condition* IE can be expressed by a list of subcounters to be measured (i.e. as a list of labels), or by a list of test conditions that need to be passed, or by a combination of both.

The rest of the subscription information follows the same as described in 7.4.2.2.

7.4.4.3 REPORT Service *RIC Indication Header* IE contents

REPORT Service style 3 shall use the *RIC Indication Header* IE Format 1 (see clause 8.2.1.3.1) as described in 7.4.2.3.

7.4.4.4 REPORT Service *RIC Indication Message* IE contents

REPORT Service style 3 shall use the *RIC Indication Message* IE Format 2 (see clause 8.2.1.4.2).

REPORT Service *RIC Indication Message* IE Format 2 carries a set of UE-level measurement data matching subscribed conditions. The included *Measurement Information Condition UE List* IE indicates the order of measured values for each measurement record in the reported data – a list of all the UE ID(s) satisfying the subscribed *Matching Condition* IE for each requested measurement within a Reporting Period.

In every granularity period during which a UE matching a subscribed condition stays in the E2 Node and maintains the RRC_CONNECTED or RRC_INACTIVE state, the E2 Node shall collect the related data and reports it at the end of the reporting period.

The *List of matched UE IDs* IE for a certain measurement type in the *Measurement Information Condition UE List* IE indicates all the UE ID(s) that satisfied the subscribed *Matching Condition* IE for that measurement type and maintained the RRC_CONNECTED or RRC_INACTIVE state at least for one granularity period during the reporting period. The *List of matched UE IDs* IE can be omitted for a certain subscribed measurement type if none of the UEs were matched during the reporting period. If the *List of matched UE IDs* IE is used for a measurement type, then the same IE shall be used for all the measurement types in the *Measurement Information Condition UE List* IE.

If the *List of matched UE IDs* IE is used, in the granularity periods where the UE does not appear in the RRC_CONNECTED or RRC_INACTIVE state (e.g. transitioned to RRC_IDLE or UE identity track is lost), the E2 Node shall not collect the related data and NULL shall be reported for those granularity periods until the end of the Reporting Period. In this case, the E2 Node shall stop reporting measurements related to this UE in the subsequent reporting periods. If the *List of matched UE IDs* IE is used and a UE whose ID is included in the IE appeared in the middle of the reporting period, then NULL shall be reported for the granularity periods priori to the UE appearing in the E2 Node.

On the other hand, the *Sequence of Matched UE IDs for Granularity Periods* IE for a certain measurement type in the *Measurement Information Condition UE List* IE can be used to indicate the UE ID(s) that satisfied the subscribed *Matching Condition* IE for the corresponding measurement type and maintained the RRC_CONNECTED or RRC_INACTIVE state, separately for each and every granularity period in chronological order. If the *Sequence of Matched UE IDs for Granularity Periods* IE is used for a measurement type, then the same IE shall be used for all the measurement types in the *Measurement Information Condition UE List* IE, and the *List of matched UE IDs* IE, if included for any measurement type, shall be ignored.

The rest of the information follows the same as described in 7.4.2.4.

7.4.5 REPORT Service Style 4: Common condition-based, UE-level Measurement

7.4.5.1 REPORT Service Style description

REPORT Service style 4 provides the UE-level performance measurement information collection for a group of UEs across a set of measurement types matching common subscribed conditions from an E2 Node.

7.4.5.2 REPORT Service *RIC Action Definition* IE contents

REPORT Service style 4 shall use the *RIC Action Definition* IE Format 4 (see clause 8.2.1.2.4), where a *Matching Condition* IE serves as a condition to include the matched UEs' measurement values into the reporting, common for each requested measurement within the *Measurement Information List* IE. The *Matching Condition* IE is expressed by a list of test conditions to filter matching UEs.

The rest of the subscription information follows the same as described in 7.4.2.2.

7.4.5.3 REPORT Service *RIC Indication Header* IE contents

REPORT Service style 4 shall use the *RIC Indication Header* IE Format 1 (see clause 8.2.1.3.1) as described in 7.4.2.3.

7.4.5.4 REPORT Service *RIC Indication Message* IE contents

REPORT Service style 4 shall use the *RIC Indication Message* IE Format 3 (see clause 8.2.1.4.3).

REPORT Service *RIC Indication Message* IE Format 3 carries a list of measurement data for UE(s) matching subscribed conditions.

In every granularity period during which a UE matching a subscribed condition stays in the E2 Node and maintains the RRC_CONNECTED or RRC_INACTIVE state, the E2 Node shall collect the related data and report it at the end of the reporting period. In the granularity periods where the UE does not appear in the RRC_CONNECTED or RRC_INACTIVE state (e.g. transitioned to RRC_IDLE or UE identity track is lost), the E2 Node shall not collect the related data and NULL shall be reported for those granularity periods until the end of the Reporting Period. In this case, the E2 Node shall stop reporting measurements related to this UE in the subsequent reporting periods.

If none of the UEs were matched during the reporting period, then E2 Node shall not report measurements for that reporting period.

The rest of the information follows the same as described in 7.4.2.4.

7.4.6 REPORT Service Style 5: E2 Node Measurement for Multiple UEs

7.4.6.1 REPORT Service Style description

REPORT Service style 5 provides the performance measurement information collection for multiple UEs of interest from an E2 Node.

7.4.6.2 REPORT Service *RIC Action Definition* IE contents

REPORT Service style 5 shall use the *RIC Action Definition* IE Format 5 (see clause 8.2.1.2.5), where the included UE Identifiers indicates UEs of interest for measurement collection.

The rest of the subscription information follows the same as described in 7.4.2.2.

7.4.6.3 REPORT Service *RIC Indication Header* IE contents

REPORT Service style 5 shall use the *RIC Indication Header* IE Format 1 (see clause 8.2.1.3.1) as described in 7.4.2.3.

7.4.6.4 REPORT Service *RIC Indication Message* IE contents

REPORT Service style 5 shall use the *RIC Indication Message* IE Format 3 (see clause 8.2.1.4.3) as described in 7.4.5.4, where the measurement data reported is associated to multiple UEs that was subscribed and available in the system.

7.4.7 REPORT Service Style 255: Multiple report measurements

7.4.7.1 REPORT Service Style description

REPORT Service style 255 provides support for the integration of multiple report jobs into a single report action from an E2 Node.

This REPORT Service style provides a mechanism to initiate multiple report action jobs of the selected fundamental level REPORT Service style(s) that should be processed in an integrated manner by the E2 Node, i.e. the RIC Subscription procedure is considered failed if at least one of the indicated report actions is unsuccessfully subscribed, and RIC SUBSCRIPTION FAILURE message shall be sent containing the appropriate error cause.

The resulting measurement reports may be either transmitted as an integrated single Report message consisting of a list of individual job specific Report messages or as individual Report messages each corresponding to a single job in the list defined in the *RIC Action Definition* IE.

7.4.7.2 REPORT Service *RIC Action Definition* IE contents

REPORT Service style 255 shall use *E2SM-KPM Action Definition Format 255* IE (see clause 8.2.1.2.6).

This REPORT Service *RIC Action Definition* IE contains a list of jobs, each associated with a unique *Job ID* IE. For each requested job, the *Job Specific Action Definition* IE corresponding to one of the Action Definition Formats of the fundamental level REPORT Services may be included. The *Common Action Definition* IE, if included, shall apply for a job where the corresponding *Job Specific Action Definition* IE is absent. In other words, the *Common Action Definition* IE shall be included, if at least one job does not have the corresponding *Job Specific Action Definition* IE configured. In this case, one or more optional job specific configuration IEs (*Job Specific Cell Global ID* IE, *Job Specific UE ID* IE, *Job Specific List of Subscribed UE IDs* IE) may also be included for such job, depending on which Action Definition Format of the fundamental level REPORT Services is configured for the *Common Action Definition* IE.

The following table presents the applicable report styles for each job specific IEs:

Job specific IE	**Applicable Fundamental level Report Style**	**E2SM-KPM Action Definition Format**	**Applicable IE**
Job Specific Cell Global ID IE	1	E2SM-KPM Action Definition Format 1	*Cell Global ID* IE
	2	E2SM-KPM Action Definition Format 1, contained within *Subscription Information* IE in E2SM-KPM Action Definition Format 2	*Cell Global ID* IE
	3	E2SM-KPM Action Definition Format 3	*Cell Global ID* IE
	4	E2SM-KPM Action Definition Format 1, contained within *Subscription Information* IE in E2SM-KPM Action Definition Format 4	*Cell Global ID* IE
	5	E2SM-KPM Action Definition Format 1, contained within *Subscription Information* IE in E2SM-KPM Action Definition Format 5	*Cell Global ID* IE
Job Specific UE ID IE	2	E2SM-KPM Action Definition Format 2	*UE ID* IE
Job Specific List of Subscribed UE IDs IE	5	E2SM-KPM Action Definition Format 5	*List of Subscribed UE IDs* IE

This REPORT Service *RIC Action Definition* IE may also include the *Reporting Approach* IE to indicate if the resulting RIC Indication message shall carry a single report integrating multiple job results, or if each job result should be reported independently via a separate RIC Indication message at each event trigger.

Example The *Common Action Definition* IE corresponding to the E2SM-KPM Action Definition Format 1 is included, with different *Job Specific Cell Global ID* IE for each requested job in the *List of Multi Report Action Definitions* IE. In this case, each job is initiated according to the fundamental level REPORT Service Style 1, with the *Cell Global ID* IE replaced by the *Job Specific Cell Global ID* IE configured for each job.

7.4.7.3 REPORT Service *RIC Indication Header* IE contents

REPORT Service style 255 shall use the *RIC Indication Header* IE Format 1 (see clause 8.2.1.3.1) as described in 7.4.2.3.

The *Job ID* IE shall be included if the *Reporting Approach* IE is included in this REPORT Service *RIC Action Definition* IE and set to “Single Integrated”.

7.4.7.4 REPORT Service *RIC Indication Message* IE contents

REPORT Service style 255 shall use the *E2SM-KPM Indication Message* IE Format 255 (see clause 8.2.1.4.4) if the *Reporting Approach* IE is included in this REPORT Service *RIC Action Definition* IE and set to “Single Integrated”. In this case, the message carries a sequence of *RIC Indication Message* IE for each job identified by the *Job ID* IE, carrying the job result formatted according to the associated fundamental level REPORT Service specific *RIC Indication Message* IE.

If the *Reporting Approach* IE is included in this REPORT Service *RIC Action Definition* IE and set to “Multiple Individually”, the *E2SM-KPM Indication Message* IE Format 255 shall not be used. For each job result, an individual RIC Indication message shall be sent, formatted according to the associated fundamental level REPORT Service specific *RIC Indication Message* IE.

In all cases, the individual measurements and measurement records are ordered according to the definition in the applicable fundamental level REPORT Service style.

7.5 Supported RIC INSERT Service Styles

Void

7.6 Supported RIC CONTROL Service Styles

Void.

7.7 Supported RIC POLICY Service Styles

Void.

7.7A Supported RIC QUERY Service Styles

Void.

7.8 Supported RIC Styles and E2SM IE Formats

Table 7.8-1 and 7.8-2 provide a summary of the E2SM IE Formats defined to support this E2SM specification.

Table 7.8-1: Summary of the E2SM IE Formats defined to support RIC Event Trigger Styles

RIC Event Trigger Style	Event Trigger Definition Format
Style 1	1
Style 2	2

Table 7.8-2: Summary of the E2SM IE Formats defined to support RIC Service Styles

RIC Service Style	Action Definition Format	Indication Header Format	Indication Message Format	Call Process ID Format	Control Header Format	Control Message Format	Control Outcome Format
REPORT							
Style 1	1	1	1				
Style 2	2	1	1				
Style 3	3	1	2				
Style 4	4	1	3				
Style 5	5	1	3				
Style 255	255	1	255				
INSERT							
CONTROL							
POLICY							
QUERY							

7.9 Conversion of measurements derived from 3GPP defined measured values

7.9.0 Conversion of measurement definitions

UE level and QoS Flow level measurements are defined based on the conversion of the measurements' definitions provided in TS 28.552 [4], TS 32.425 [8], and O-RAN specific measurement defined in clause 7.10 according to the following rules:

The type of the original measurements	The corresponding per-UE and per-UE-per-slice measurements	The corresponding per-QoS flow and per-slice-per-QoS flow measurements	Notes
Throughput Delay Data volume In-session activity time	Measured per UE	Measured per QoS flow	For the Throughput and Data volume measurements, the formulas specified in 3GPP are used with restriction to the individual UE or individual QoS flow , and also based on clause 7.9.1.
PDCP drop rate IP latency	Measured per UE	Measured per QoS flow	For the Throughput and Data volume measurements, the formulas specified in 3GPP are used with restriction to the individual UE or individual QoS flow.
Radio resource utilization	Measured per UE	N/A	The formulas specified in 3GPP are used with restriction to the individual UE.
RRC connections related PDU sessions related DRBs related QoS flows related	Measured per UE	N/A	
Mobility management	Measured per UE	N/A	
CQI related MCS related	Measured per UE	N/A	
PEE related	N/A	N/A	

Distribution of Normally/Abnormally Released Calls	Measured per UE	N/A	

Beam level measurements are defined based on the conversion of the measurements' definitions provided in TS 28.552 [4], TS 32.425 [8], and O-RAN specific measurement defined in clause 7.10 according to the following rules:

The type of the original measurements	The corresponding per-beam measurements	Notes
Mobility Management	Measured per source beam and neighboring cell pair	The formulas specified in 3GPP are used with restriction to the source beam-neighboring cell pair.

7.9.1 Changes in the units of measurements while adopting for E2SM-KPM

The units of the following measurements in TS 28.552 [4] and TS 32.425 [8] are replaced with newer units, as shown in the table below.

Measurement Type	Measurement Name	Data Type	Unit used in 3GPP	Unit used in E2SM-KPM
DL Cell PDCP SDU Data Volume, defined in TS 28.552 [4] clause 5.1.2.1.2.1.	DRB.PdcpSduVolumeDL_Filter	INTEGER	Mbit	Kbit
UL Cell PDCP SDU Data Volume, defined in TS 28.552 [4] clause 5.1.2.1.2.2.	DRB.PdcpSduVolumeUL_Filter	INTEGER	Mbit	Kbit
DL PDCP PDU Data Volume, defined in TS 28.552 [4] clause 5.1.3.6.1.1.	QosFlow.PdcpPduVolumeDL_Filter	INTEGER	Mbit	Kbit
UL PDCP PDU Data Volume, defined in TS 28.552 [4] clause 5.1.3.6.1.2.	QosFlow.PdcpPduVolumeUL_Filter	INTEGER	Mbit	Kbit
DL PDCP SDU Data Volume, defined in TS 28.552 [4] clause 5.1.3.6.2.1.	QosFlow.PdcpSduVolumeDl_Filter	INTEGER	Mbit	Kbit
UL PDCP SDU Data Volume, defined in TS 28.552 [4] clause 5.1.3.6.2.2.	QosFlow.PdcpSduVolumeUl_Filter	INTEGER	Mbit	Kbit
DL Cell PDCP SDU Data Volume on X2 interface, defined in TS 28.552 [4] clause 5.1.2.1.1.2.	DRB.PdcpSduVolumeX2DL_Filter	INTEGER	Mbit	Kbit
UL Cell PDCP SDU Data Volume on X2 interface, defined in TS 28.552 [4] clause 5.1.2.1.2.2.	DRB.PdcpSduVolumeX2UL_Filter	INTEGER	Mbit	Kbit
DL Cell PDCP SDU Data Volume on Xn interface, defined in TS 28.552 [4] clause 5.1.2.1.1.3.	DRB.PdcpSduVolumeXnDL_Filter	INTEGER	Mbit	Kbit
UL Cell PDCP PDU Data Volume on Xn interface, defined in TS 28.552 [4] clause 5.1.2.1.2.3.	DRB.PdcpSduVolumeXnUL_Filter	INTEGER	Mbit	Kbit
DL PDCP SDU Data Volume per interface, defined in TS 28.552 [4] clause 5.1.3.6.2.3.	DRB.F1uPdcpSduVolumeDl.*QoS*, DRB.X2uPdcpSduVolumeDl.*QoS*, DRB.XnuPdcpSduVolumeDl.*QoS*	INTEGER	Mbit	Kbit
UL PDCP SDU Data Volume per interface, defined in TS 28.552 [4] clause 5.1.3.6.2.4.	DRB.F1uPdcpSduVolumeUl.*QoS*, DRB.X2uPdcpSduVolumeUl.*QoS*, DRB.XnuPdcpSduVolumeUl.*QoS*	INTEGER	Mbit	Kbit
DL cell PDCP SDU Data Volume, defined in TS 32.425 [8] clause 4.4.7.1.	DRB.PdcpSduVolumeDl_Filter	INTEGER	Mbit	Kbit
UL cell PDCP SDU Data Volume, defined in TS 32.425 [8] clause 4.4.7.2.	DRB.PdcpSduVolumeUl_Filter	INTEGER	Mbit	Kbit
In-session activity time for UE, defined in TS 28.552 [4] clause 5.1.1.13.2.2.	QF.SessionTimeUE	INTEGER	s	ms
In-session activity time for DRB, defined in TS 28.552 [4] clause 5.1.1.10.4.	DRB.SessionTime.5QI, DRB.SessionTime.SNSSAI	INTEGER	s	ms
In-session activity time for QoS flow, defined in TS 28.552 [4] clause 5.1.1.13.2.1.	QF.SessionTimeQoS.*QoS*	INTEGER	s	ms
In-session activity time for UE, defined in TS 32.425 [8] clause 4.2.4.1.	ERAB.SessionTimeUE	INTEGER	s	ms
In-session activity time for E-RABs, defined in TS 32.425 [8] clause 4.2.4.2.	ERAB.SessionTimeQCI.*QCI*	INTEGER	s	ms
IP throughput in UL, defined in TS 32.425 [8] clause 4.4.6.2.	DRB.IPThpUl.*QCI*	REAL	Kbit	Kbit/s

The changes in the units of the measurements shown in the above table are to prevent the reported values from being reported as 0 caused by rounding off the precision in the decimals to report them as INTEGER, except the last row of “IP throughput in UL”, which is to fix the erroneous unit.

7.10 O-RAN specific Performance Measurement

7.10.1 DL Transmitted Data Volume (RLC SDU)

Measurement Name	**DL Transmitted Data Volume (RLC SDU)**
a) Description	This measurement provides the transmitted data volume in the downlink in a measurement time. The measurement is split into subcounters per QoS level (mapped 5QI or QCI in EN-DC), and subcounters per supported S-NSSAI. The unit is kbit.
b) Collection Method	CC
c) Condition	This measurement is obtained by counting the data volume counted on RLC SDU level, in kbit successfully transmitted (acknowledged by UE) in DL for one DRB during measurement time T. Separate counters are maintained for each mapped 5QI (or QCI for option 3) and for each supported S-NSSAI.
d) Measurement Result	Each measurement is an integer value representing the number of bits measured in kbits (1kbits=1000 bits). The number of measurements is equal to the number of PLMNs multiplied by the number of QoS levels multiplied by the number of S-NSSAIs.
e) Measurement Type	The measurement name has the form DRB.RlcSduTransmittedVolumeDL_Filter. Where filter is a combination of PLMN ID and QoS level and S-NSSAI. Where PLMN ID represents the PLMN ID, QoS represents the mapped 5QI or the QCI level, and SNSSAI represents S-NSSAI.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for performance assurance within integrity area (user plane connection quality).

7.10.2 UL Transmitted Data Volume (RLC SDU)

Measurement Name	**UL Transmitted Data Volume (RLC SDU)**
a) Description	This measurement provides the transmitted data volume in the uplink in a certain period. The measurement is split into subcounters per QoS level (mapped 5QI or QCI in EN-DC), and subcounters per supported S-NSSAI. The unit is kbit.
b) Collection Method	CC
c) Condition	This measurement is obtained by counting the data volume counted on RLC SDU level, in kbit successfully transmitted (acknowledged by E2 Node) in UL for one DRB during measurement time T. Separate counters are maintained for each mapped 5QI (or QCI for option 3) and for each supported S-NSSAI.
d) Measurement Result	Each measurement is an integer value representing the number of bits measured in kbits (1kbits=1000 bits). The number of measurements is equal to the number of PLMNs multiplied by the number of QoS levels multiplied by the number of S-NSSAIs.
e) Measurement Type	The measurement name has the form DRB.RlcSduTransmittedVolumeUL_Filter. Where filter is a combination of PLMN ID and QoS level and S-NSSAI. Where PLMN ID represents the PLMN ID, QoS represents the mapped 5QI or the QCI level, and SNSSAI represents S-NSSAI.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for performance assurance within integrity area (user plane connection quality).

7.10.3 Distribution of Percentage of DL Transmitted Data Volume to Incoming Data Volume

Measurement Name	**Distribution of Percentage of DL Transmitted Data Volume to Incoming Data Volume**
a) Description	This measurement provides the distribution of the percentage of successfully transmitted data volume to incoming data volume in downlink for UEs. The measurement is split into subcounters per QoS level (mapped 5QI or QCI in EN-DC), and subcounters per supported S-NSSAI.
b) Collection Method	CC
c) Condition	The Measurement is calculated by 100*x/y for each UE. x is incremented by counting the number of bits counted on RLC SDU level successfully transmitted (acknowledged by UE) in DL for one DRB during measurement time T. y is incremented by counting the number of bits entering the RLC layers in DL for one DRB during measurement time T. For each UE, the bin corresponding to the percentage of transmitted data volume to incoming data volume (100*x/y) experienced by the UE is incremented by one. Separate counters are maintained for each mapped 5QI (or QCI for option 3) and for each supported S-NSSAI.
d) Measurement Result	A set of integers, each representing the (integer) number of samples with a percentage of DL transmitted data volume to incoming data volume in the range represented by that bin. If the optional QoS level subcounter and S-NSSAI subcounter and PLMN ID subcounter measurements are performed, the number of measurements is equal to the number of mapped 5QIs and the number of supported S-NSSAIs, and the number of PLMN IDs.
e) Measurement Type	The measurement name has the form DRB.PerDataVolumeDLDist.Bin where Bin represents the bin, or optionally DRB.PerDataVolumeDLDist.Bin.QOS, where QOS identifies the target quality of service class, and DRB.PerDataVolumeDLDist.Bin.SNSSAI, where SNSSAI identifies the S-NSSAI, and DRB.PerDataVolumeDLDist.Bin.PLMN, where PLMN identifies the PLMN ID.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Packet Switched
h) Generation	5GS
i) Purpose	Network Operator’s Traffic Engineering Community

7.10.4 Distribution of Percentage of UL Transmitted Data Volume to Incoming Data Volume

Measurement Name	**Distribution of Percentage of UL Transmitted Data Volume to Incoming Data Volume**
a) Description	This measurement provides the distribution of the percentage of successfully transmitted data volume to incoming data volume in uplink for UEs. The measurement is split into subcounters per QoS level (mapped 5QI or QCI in EN-DC), and subcounters per supported S-NSSAI.
b) Collection Method	CC
c) Condition	The Measurement is calculated by 100*x/y for each UE. x is incremented by counting the number of bits counted on RLC SDU level successfully transmitted (acknowledged by E2 Node) in UL for one DRB during measurement time T. y is incremented by counting the number of bits entering the RLC layers in UL for one DRB during measurement time T. It is up to implementation how to measure y reliably during T. For each UE, the bin corresponding to the percentage of transmitted data volume to incoming data volume (100*x/y) experienced by the UE is incremented by one. Separate counters are maintained for each mapped 5QI (or QCI for option 3) and for each supported S-NSSAI.
d) Measurement Result	A set of integers, each representing the (integer) number of samples with a percentage of UL transmitted data volume to incoming data volume in the range represented by that bin. If the optional QoS level subcounter and S-NSSAI subcounter and PLMN ID subcounter measurements are performed, the number of measurements is equal to the number of mapped 5QIs and the number of supported S-NSSAIs, and the number of PLMN IDs.
e) Measurement Type	The measurement name has the form DRB.PerDataVolumeULDist.Bin where Bin represents the bin, or optionally DRB.PerDataVolumeULDist.Bin.QOS, where QOS identifies the target quality of service class, and DRB.PerDataVolumeULDist.Bin.SNSSAI, where SNSSAI identifies the S-NSSAI, and DRB.PerDataVolumeUlDist.Bin.PLMN, where PLMN identifies the PLMN ID.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Packet Switched
h) Generation	5GS
i) Purpose	Network Operator’s Traffic Engineering Community

7.10.5 Distribution of DL Packet Drop Rate

Measurement Name	**Distribution of DL Packet Drop Rate**
a) Description	This measurement provides the fraction of RLC SDU packets which are dropped on the downlink, due to high traffic load, traffic management etc in the gNB-DU. Only user-plane traffic (DTCH) is considered. A dropped packet is one without any part of it having been transmitted on the air interface. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC), and subcounters per supported S-NSSAI.
b) Collection Method	SI
c) Condition	This attribute is created by counting the number of UEs experiencing a certain packet loss rate in each range.
d) Measurement Result	Each measurement is an integer value representing the drop rate multiplied by 1E6 of each UE within the range of the bin. The number of measurements is equal to one. If the optional QoS and S-NSSAI level measurement are performed, the measurements are equal to the number of mapped 5QIs and the number of supported S-NSSAIs.
e) Measurement Type	The measurement name has the form DRB.RlcPacketDropRateDLDist and optionally DRB.RlcPacketDropRateDLDist.QOS where QOS identifies the target quality of service class, and DRB.RlcPacketDropRateDLDist.SNSSAI where SNSSAI identifies the S-NSSAI.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for performance assurance within integrity area (user plane connection quality).

7.10.6 Distribution of UL Packet Loss Rate

Measurement Name	Distribution of UL Packet Loss Rate
a) Description	This measurement provides the distribution of the fraction of PDCP SDU packets which are not successfully received at gNB-CU-UP. It is a measure of the UL packet loss including any packet losses in the air interface, in the gNB-CU and on the F1-U interface. Only user-plane traffic (DTCH) and only PDCP SDUs that have entered PDCP (and given a PDCP sequence number) are considered. The measurement is optionally split into subcounters per QoS level (mapped 5QI or QCI in EN-DC), and subcounters per supported S-NSSAI.
b) Collection Method	SI
c) Condition	This attribute is created by counting the number of UEs experiencing a certain packet loss rate in each range.
d) Measurement Result	Each measurement is an integer value representing the loss rate multiplied by 1E6 of each UE within the range of the bin. If the optional QoS and S-NSSAI level measurement are performed, the measurements are equal to the number of mapped 5QIs and the number of supported S-NSSAIs.
e) Measurement Type	The measurement name has the form DRB.PacketLossRateULDist and optionally DRB.PacketLossRateULDist.QOS where QOS identifies the target quality of service class, and DRB. PacketLossRateULDist.SNSSAI where SNSSAI identifies the S-NSSAI.
f) Measurement Object Class	GNBCUUPFunction. NRCellCU.
g) Switching Technology	Valid for packet switched traffic.
h) Generation	5GS.
i) Purpose	One usage of this measurement is for performance assurance within integrity area (user plane connection quality).

7.10.7 DL Synchronization Signal based Reference Signal Received Power (SS-RSRP)

Measurement Name	**DL Synchronization Signal based Reference Signal Received Power (SS-RSRP)**
a) Description	This measurement provides the average of the DL SS-RSRP (see TS 38.215 [16]) values reported from UEs in the cell when SS-RSRP is used for L1-RSRP as configured by reporting configurations as defined in TS 38.214 [18], in case the L1-RSRP report function is enabled. Separate counters are maintained for each SSB in the cell.
b) Collection Method	DER (N=1)
c) Condition	This measurement is obtained by taking the average of the reported DL SS-RSRP values (i.e. between RSRP_0 and RSRP_126, see Table 10.1.6.1-1 in TS 38.133 [17]) from UEs in the cell per SSB during a granularity period.
d) Measurement Result	Each counter is an real value representing the average of the reported DL SS-RSRP values (i.e. between RSRP_0 and RSRP_126, see Table 10.1.6.1-1 in TS 38.133 [17]) for each SSB. The number of measurements is equal to the number of SSB beams defined in the cell.
e) Measurement Type	The measurement name has the form L1M.DL-SS-RSRP.*SSB*, where *SSB* represents the counter associated with SSB.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for mMIMO Non-GoB optimization in [21].

7.10.8 DL Synchronization Signal based Signal to Noise and Interference Ratio (SS-SINR)

Measurement Name	**DL Synchronization Signal based Signal to Noise and Interference Ratio (SS-SINR)**
a) Description	This measurement provides the average of the DL SS-SINR (see TS 38.215 [16]) values reported from UEs in the cell when SS-SINR is used for L1-SINR as configured by reporting configurations as defined in TS 38.214 [18], in case the L1-SINR report function is enabled. Separate counters are maintained for each SSB in the cell.
b) Collection Method	DER (N=1)
c) Condition	This measurement is obtained by taking the average of the reported DL SS-SINR values (i.e. between SINR_0 and SINR_127, see Table 10.1.16.1-1 in TS 38.133 [17]) from UEs in the cell per SSB during a granularity period.
d) Measurement Result	Each counter is an real value representing the average of the reported DL SS-SINR values (i.e. between SINR_0 and SINR_127, see Table 10.1.16.1-1 in TS 38.133 [17]) for each SSB. The number of measurements is equal to the number of SSB beams defined in the cell.
e) Measurement Type	The measurement name has the form L1M.DL-SS-SINR.*SSB*, where *SSB* represents the counter associated with SSB.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for mMIMO Non-GoB optimization in [21].

7.10.9 UL Sounding Reference Signal based Reference Signal Received Power (SRS-RSRP)

Measurement Name	**UL Sounding Reference Signal based Reference Signal Received Power (SRS-RSRP)**
a) Description	This measurement provides the average of the UL SRS-RSRP (see TS 38.215 [16]) values measured for UEs in the cell.
b) Collection Method	DER (N=1)
c) Condition	This measurement is obtained by taking the average of the measured UL SRS-RSRP values for UEs in the cell during a granularity period.
d) Measurement Result	The measurement is an real value representing the average of the measured UL SRS-RSRP values.
e) Measurement Type	The measurement name has the form L1M.UL-SRS-RSRP.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for mMIMO Non-GoB optimization in [21].

7.10.10 Total number of scheduled time slots

7.10.10.1 Cell-specific DL Total transmission time duration

Measurement Name	**Cell-specific DL total transmission time duration**
a) Description	This measurement provides the total duration of time that covers the total number of time slots within each measurement granularity period, when one or more transport blocks were scheduled for HARQ downlink transmission (including HARQ retransmission) from the radio resources of a given component carrier (i.e., a given cell) to the UEs. The total number of time slots scheduled for a UE includes the unique time slots scheduled from the component carrier, irrespective of whether it is the primary component carrier (i.e., the primary cell) or a secondary supplemental component carrier (i.e., the secondary cell) to the UE, for downlink transmission of one or more transport blocks to the UE. The unit is in micro-second [μs]. The measurement is optionally split into sub-counters per QoS level (mapped 5QI or QCI in EN-DC) and sub-counters per supported S-NSSAI, and sub-counters per PLMN ID, and sub-counters from BWP.
b) Collection Method	CC
c) Condition	The counter is incremented by the length of the time slot (in micro-seconds) during every time slot, when a HARQ downlink transmission is scheduled on the radio resources of the cell to one or more UEs served by the cell, either in its capacity as the primary cell or a secondary cell for the UEs. For this counter to be generated at a per-UE level, the counter is incremented by the length of the time slot (in micro-seconds), during every time slot when a HARQ transmission is scheduled on the radio resources of the cell to the given UE served by the cell, either in its capacity as the primary cell or a secondary cell for the UE.
d) Measurement Result	Each measurement is a single integer value. If optional measurements with sub-counters are performed, the number of measurements is equal to the number of supported QoS levels, the number of supported S-NSSAIs, the number of supported PLMNs, and the number of supported BWPs.
e) Measurement Type	The measurement name has the form: ScheduledTxTimeDl (as the default measurement, without sub-counters or filters), or optionally, ScheduledTxTimeDl.*QOS*, where *QOS* identifies the target quality of service class, ScheduledTxTimeDl.*SNSSAI* , where *SNSSAI* identifies the S-NSSAI, and ScheduledTxTimeDl.*PLMN*, where *PLMN* refers to the PLMN ID, and ScheduledTxTimeDl.*BWP,* where BWP identifies the active BWP.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for performance assurance within integrity area (user plane connection quality).

7.10.10.2 Cell-specific downlink total failed transmission time duration

Measurement Name	**Cell-specific DL total failed transmission time duration**
a) Description	This measurement provides the total duration of time that covers the total number of time slots within each measurement granularity period, when one or more transport blocks were unsuccessfully transmitted in the downlink from the radio resources of a given component carrier (i.e., a given cell) to the UEs, resulting in HARQ NACKs from the UEs. For a given UE, this measurement indicates the total number of unique time slots scheduled from the component carrier, irrespective of whether it is the primary component carrier (i.e., the primary cell) or a secondary supplemental component carrier (i.e., the secondary cell) to the UE, which resulted in unsuccessful transmission of one or more transport blocks to the UE. The unit is In micro-second [μs]. The measurement is optionally split into sub-counters per QoS level (mapped 5QI or QCI in EN-DC) and sub-counters per supported S-NSSAI, and sub-counters per PLMN ID, and sub-counters from BWP.
b) Collection Method	CC
c) Condition	The counter is incremented by the length of the time slot (in micro-seconds) during every time slot, when the O-DU receives a NACK from any UE for a HARQ downlink transmission from the radio resources of the cell to the UE, irrespective of whether the cell serves the UE as the primary component carrier (i.e., the primary cell) or a secondary supplemental component carrier (i.e., the secondary cell). For this counter to be generated at a per-UE level, the counter is incremented by the length of the time slot (in micro-seconds) during every time slot, when the O-DU receives a NACK from the UE for a HARQ downlink transmission from the radio resources of the cell to the UE, irrespective of whether the cell serves the UE as the primary component carrier (i.e., the primary cell) or a secondary supplemental component carrier (i.e., the secondary cell).
d) Measurement Result	Each measurement is a single integer value. If optional measurements with sub-counters are performed, the number of measurements is equal to the number of supported QoS levels, the number of supported S-NSSAIs, the number of supported PLMNs, and the number of supported BWPs.
e) Measurement Type	The measurement name has the form: FailedTxTimeDl (as the default measurement, without sub-counters or filters), or optionally, FailedTxTimeDl.*QOS*, where *QOS* identifies the target quality of service class, FailedTxTimeDl.*SNSSAI*, where *SNSSAI* identifies the S-NSSAI, and FailedTxTimeDl.*PLMN*, where *PLMN* refers to the PLMN ID, and FailedTxTimeDl.*BWP,* where BWP identifies the active BWP.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for performance assurance within integrity area (user plane connection quality).

7.10.10.3 Cell-specific downlink retransmission time duration

Measurement Name	Cell-specific DL retransmission time duration
a) Description	This measurement provides the total duration of time covering the total number of time slots within each measurement granularity period, when one or more previously-unsuccessfully-transmitted transport blocks were scheduled for HARQ retransmission in the downlink from the radio resources of a given component carrier (i.e., a given cell) to the UEs. The total number of time slots scheduled for a UE includes the unique time slots scheduled from the component carrier in the downlink, irrespective of whether it is the primary component carrier (i.e., the primary cell) or a secondary supplemental component carrier (i.e., the secondary cell) to the UE, for HARQ retransmission of one or more transport blocks to the UE. The unit is in micro-second [μs]. The measurement is optionally split into sub-counters per QoS level (mapped 5QI or QCI in EN-DC) and sub-counters per supported S-NSSAI, and sub-counters per PLMN ID, and sub-counters from BWP.
b) Collection Method	CC
c) Condition	The counter is incremented by the length of the time slot (in micro-seconds) during every time slot, when a HARQ downlink re-transmission of one or more previously-unsuccessfully-transmitted transport blocks is scheduled on the radio resources of the cell to one or more UEs served by the cell, either in its capacity as the primary cell or a supplemental secondary cell for the UEs. For this counter to be generated at a per-UE level, the counter is incremented by the length of the time slot (in micro-seconds), during every time slot when a HARQ downlink re-transmission of one or more previously-unsuccessfully-transmitted transport blocks is scheduled on the radio resources of the cell to the given UE served by the cell, either in its capacity as the primary cell or a supplemental secondary cell for the UE.
d) Measurement Result	Each measurement is a single integer value. If optional measurements with sub-counters are performed, the number of measurements is equal to the number of supported QoS levels, the number of supported S-NSSAIs, the number of supported PLMNs, and the number of supported BWPs.
e) Measurement Type	The measurement name has the form: ScheduledReTxTimeDl(as the default measurement, without sub-counters or filters), or optionally, ScheduledReTxTimeDl.*QOS* , where *QOS* identifies the target quality of service class, ScheduledReTxTimeDl.*SNSSAI* where *SNSSAI* identifies the S-NSSAI, and ScheduledReTxTimeDl.*PLMN*, where *PLMN* refers to the PLMN ID, andScheduledReTxTimeDl.*BWP ,* where BWP identifies the active BWP.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for performance assurance within integrity area (user plane connection quality).

7.10.10.4 Cell-specific downlink total SU-MIMO transmission time duration

Measurement Name	**Cell-specific DL total SU-MIMO transmission time duration**
a) Description	This measurement provides the total duration of time that covers the total number of time slots within each measurement granularity period, when multiple transport blocks were scheduled for HARQ downlink transmission in SU-MIMO mode from the radio resources of a given component carrier (i.e., a given cell) to the UEs. The total number of time slots scheduled for a UE includes the unique time slots scheduled from the component carrier, irrespective of whether it is the primary component carrier (i.e., the primary cell) or a secondary supplemental component carrier (i.e., the secondary cell) to the UE, for transmission of multiple transport blocks in SU-MIMO mode to the UE. The unit is in micro-second [μs]. The measurement is optionally split into sub-counters per QoS level (mapped 5QI or QCI in EN-DC) and sub-counters per supported S-NSSAI, and sub-counters per PLMN ID, and sub-counters from BWP.
b) Collection Method	CC
c) Condition	The counter is incremented by the length of the time slot (in micro-seconds) during every time slot, when a HARQ downlink transmission of multiple transport blocks in SU-MIMO mode is scheduled on the radio resources of the cell to one or more UEs served by the cell, either in its capacity as the primary cell or a secondary cell for the UEs. For this counter to be generated at a per-UE level, the counter is incremented by the length of the time slot (in micro-seconds), during every time slot when a HARQ downlink transmission of multiple transport blocks in SU-MIMO mode is scheduled on the radio resources of the cell to the given UE served by the cell, either in its capacity as the primary cell or a secondary cell for the UE.
d) Measurement Result	Each measurement is a single integer value. If optional measurements with sub-counters are performed, the number of measurements is equal to the number of supported QoS levels, the number of supported S-NSSAIs, the number of supported PLMNs, and the number of supported BWPs.
e) Measurement Type	The measurement name has the form: ScheduledTxTimeDlSUMIMO (as the default measurement, without sub-counters or filters), or optionally, ScheduledTxTimeDlSUMIMO.*QOS* , where *QOS* identifies the target quality of service class, ScheduledTxTimeDlSUMIMO.*SNSSAI* , where *SNSSAI* identifies the S-NSSAI, and ScheduledTxTimeDlSUMIMO.*PLMN*, where *PLMN* refers to the PLMN ID, and ScheduledTxTimeDlSUMIMO.*BWP,* where BWP identifies the active BWP.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for performance assurance within integrity area (user plane connection quality).

7.10.10.5 Cell-specific downlink total MU-MIMO transmission time duration

Measurement Name	**Cell-specific DL total MU-MIMO transmission time duration**
a) Description	This measurement provides the total duration of time covering the total number of time slots within each measurement granularity period, when multiple transport blocks were scheduled for HARQ downlink transmission in MU-MIMO mode from the radio resources of a given component carrier (i.e., a given cell) to the UEs. The total number of time slots scheduled for a UE includes the unique time slots scheduled from the component carrier, irrespective of whether it is the primary component carrier (i.e., the primary cell) or a secondary supplemental component carrier (i.e., the secondary cell) to the UE, for transmission of downlink transport blocks to the UE when the UE is in MU-MIMO mode. The unit is in micro-second [μs]. The measurement is optionally split into sub-counters per QoS level (mapped 5QI or QCI in EN-DC) and sub-counters per supported S-NSSAI, and sub-counters per PLMN ID, and sub-counters from BWP.
b) Collection Method	CC
c) Condition	The counter is incremented by the length of the time slot (in micro-seconds) during every time slot, when a HARQ downlink transmission of multiple transport blocks in MU-MIMO mode is scheduled on the radio resources of the cell to multiple UEs served by the cell, either in its capacity as the primary cell or a secondary cell for the UEs. For this counter to be generated at a per-UE level, the counter is incremented by the length of the time slot (in micro-seconds), during every time slot when the UE is configured in MU-MIMO mode and when a HARQ downlink transmission of transport blocks is scheduled on the radio resources of the cell to the given UE served by the cell, either in its capacity as the primary cell or a secondary cell for the UE.
d) Measurement Result	Each measurement is a single integer value. If optional measurements with sub-counters are performed, the number of measurements is equal to the number of supported QoS levels, the number of supported S-NSSAIs, the number of supported PLMNs, and the number of supported BWPs.
e) Measurement Type	The measurement name has the form: ScheduledTxTimeDlMUMIMO (as the default measurement, without sub-counters or filters), or optionally, ScheduledTxTimeDlMUMIMO.*QOS*, where *QOS* identifies the target quality of service class, ScheduledTxTimeDlMUMIMO.*SNSSAI*, where *SNSSAI* identifies the S-NSSAI, and ScheduledTxTimeDlMUMIMO.*PLMN*, where *PLMN* refers to the PLMN ID, and ScheduledTxTimeDlMUMIMO.*BWP,* where BWP identifies the active BWP.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for performance assurance within integrity area (user plane connection quality).

7.10.11 DL Transmitted Data Volume (MAC SDU)

Measurement Name	**DL Transmitted Data Volume (MAC SDU)**
a) Description	This measurement provides the transmitted data volume in the downlink in a measurement time. The measurement is split into subcounters per PLMN ID and subcounters per QoS level (mapped 5QI or QCI in NR option 3), and subcounters per supported S-NSSAI. The unit is kbit.
b) Collection Method	CC
c) Condition	This measurement is obtained by counting the data volume counted on MAC SDU level, in kbit successfully transmitted (reception of HARQ ACK) in DL for one DRB during measurement time T. Separate counters are maintained for each mapped 5QI (or QCI for option 3) and for each supported S-NSSAI.
d) Measurement Result	Each measurement is an integer value representing the number of bits measured in kbits (1kbits=1000 bits). The number of measurements is equal to the number of PLMNs multiplied by the number of QoS levels multiplied by the number of S-NSSAIs.
e) Measurement Type	The measurement name has the form DRB.MacSduTransmittedVolumeDL_Filter. Where filter is a combination of PLMN ID and QoS level and S-NSSAI. Where PLMN ID represents the PLMN ID, QoS represents the mapped 5QI or the QCI level, and SNSSAI represents S-NSSAI.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for performance assurance within integrity area (user plane connection quality).

7.10.12 UL Transmitted Data Volume (MAC SDU)

Measurement Name	**UL Transmitted Data Volume (MAC SDU)**
a) Description	This measurement provides the transmitted data volume in the uplink in a certain period. The measurement is split into subcounters per QoS level (mapped 5QI or QCI in NR option 3), and subcounters per supported S-NSSAI. The unit is kbit.
b) Collection Method	CC
c) Condition	This measurement is obtained by counting the data volume counted on MAC SDU level, in kbit successfully transmitted (transmission of HARQ ACK) in UL for one DRB during measurement time T. Separate counters are maintained for each mapped 5QI (or QCI for option 3) and for each supported S-NSSAI.
d) Measurement Result	Each measurement is an integer value representing the number of bits measured in kbits (1kbits=1000 bits). The number of measurements is equal to the number of PLMNs multiplied by the number of QoS levels multiplied by the number of S-NSSAIs.
e) Measurement Type	The measurement name has the form DRB.MacSduTransmittedVolumeUL_Filter. Where filter is a combination of PLMN ID and QoS level and S-NSSAI. Where PLMN ID represents the PLMN ID, QoS represents the mapped 5QI or the QCI level, and SNSSAI represents S-NSSAI.
f) Measurement Object Class	NRCellDU
g) Switching Technology	Valid for packet switched traffic
h) Generation	5GS
i) Purpose	One usage of this measurement is for performance assurance within integrity area (user plane connection quality).

8 Elements for E2SM Service Model

8.1 General

Clause 8.2 describes the structure of the information elements as required for E2SM-KPM in tabular format. Clause 8.3 presents individual information elements. Clause 8.4 provides the corresponding ASN.1 definition of each information element.

The following attributes are used for the tabular description of the messages and information elements:

NOTE: The messages have been defined by the guidelines specified in 3GPP TR 25.921 [i.3].

8.2 Message Functional Definition and Content

8.2.1 Messages for RIC Functional procedures

8.2.1.1 RIC EVENT TRIGGER DEFINITION IE

This information element is part of the RIC SUBSCRIPTION REQUEST message sent by the Near-RT RIC to an E2 Node and is required for event triggers used to initiate REPORT actions.

Direction: NEAR-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Event Trigger Format*	M			
>E2SM-KPM Event Trigger Definition Format 1			8.2.1.1.1	
>E2SM-KPM Event Trigger Definition Format 2			8.2.1.1.2	

8.2.1.1.1 E2SM-KPM Event Trigger Definition Format 1

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Reporting Period	M		INTEGER (1..4294967295)	The reporting period is expressed in unit of 1 millisecond.

8.2.1.1.2 E2SM-KPM Event Trigger Definition Format 2

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Sequence of Event Trigger		*1.. <maxnoofEventTrigger>*		
>Event Condition	M		8.3.31 Event Formula	Defines the condition for event evaluation.
>Event Evaluation Period	O		INTEGER (1..4294967295)	Time interval for event evaluation, expressed in unit of 1 millisecond.
>Logical OR	O		ENUMERATED (true, …)	If set to "true", logical connection to the next condition is "or".

Range bound	Explanation
maxnoofEventTrigger	Maximum number of event trigger. Value is <65535>.

8.2.1.2 RIC ACTION DEFINITION IE

This information element is part of the RIC SUBSCRIPTION REQUEST message sent by the Near-RT RIC to an E2 Node. In this service model, this information element provides additional information for the nominated Action (Report).

Direction: NEAR-RT RIC → E2 Node.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RIC Style Type	M		8.3.3	
CHOICE *Action Definition Format*	M			
>E2SM-KPM Action Definition Format 1			8.2.1.2.1	
>E2SM-KPM Action Definition Format 2			8.2.1.2.2	
>E2SM-KPM Action Definition Format 3			8.2.1.2.3	
>E2SM-KPM Action Definition Format 4			8.2.1.2.4	
>E2SM-KPM Action Definition Format 5			8.2.1.2.5	
>E2SM-KPM Action Definition Format 255			8.2.1.2.6	Used to generate list of fundamental level E2SM-KPM Action Definition IEs

8.2.1.2.1 E2SM-KPM Action Definition Format 1

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Measurement Information List		*1..<maxnoofMeasurementInfo>*		
>CHOICE *Measurement Type*	M			
>>Measurement Name			8.3.9 Measurement Type Name	
>>Measurement ID			8.3.10 Measurement Type ID	
>List of Labels		*1..<maxnoofLabelInfo>*		
>>Label Information	M		8.3.11 Measurement Label	
>Matching Condition for Reporting		*0..<maxnoofConditionInfo>*		
>>Measured Value Reporting Condition	M		8.3.28	
>>Logical OR	O		8.3.25	
Granularity Period	M		8.3.8 Granularity Period	Collection interval of measurements
Cell Global ID	O		8.3.20 Cell Global ID	Points to a specific cell for generating measurements subscribed by the *Measurement Information List* IE This IE is ignored when Report Style 255 is used, *Common Action Definition* IE is present and for the specific job, *Job Specific Action Definition* IE is absent and *Job Specific Cell Global ID* IE is present.
Distribution Measurement Bin Range Info List		*0..<maxnoofMeasurementInfo>*		
>CHOICE Measurement Type	M			
>>Measurement Name			8.3.9 Measurement Type Name	
>>Measurement ID			8.3.10 Measurement Type ID	
>Bin Range Definition	M		8.3.26	Indicates the value ranges of bins for distribution type measurement

Range bound	Explanation
maxnoofMeasurementInfo	Maximum no. of measurement types that can be reported by a single report. Value is <65535>.
maxnoofLabelInfo	Maximum no. of measurements values that can be reported for a single measurement type. Value is <2147483647>.
maxnoofConditionInfo	Maximum no. of conditions that can be subscribed for a single measurement type. Value is <32768>.

8.2.1.2.2 E2SM-KPM Action Definition Format 2

IE/Group Name	Presence	Range	IE type and reference	Semantics description
UE ID	M		8.3.24	Points to a specific UE of interest. This IE is ignored when Report Style 255 is used, *Common Action Definition* IE is present and for the specific job, *Job Specific Action Definition* IE is absent and *Job Specific UE ID* IE is present.
Subscription Information	M		8.2.1.2.1 E2SM-KPM Action Definition Format 1	

8.2.1.2.3 E2SM-KPM Action Definition Format 3

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Measurement Information Condition List		*1..<maxnoofMeasurementInfo>*		
>CHOICE *Measurement Type*	M			
>>Measurement Name			8.3.9 Measurement Type Name	
>>Measurement ID			8.3.10 Measurement Type ID	
>Matching Condition		*1..<maxnoofConditionInfo>*		The Matching Condition represents the Boolean expression, the logical connection to the next condition is AND if the *Logical OR* IE is not included
>>CHOICE *Condition Type*	M			
>>>Label Information			8.3.11 Measurement Label	
>>>Test Information			8.3.22 Test Condition Information	
>>Logical OR	O		8.3.25	If included, logical connection to the next condition is “or”.
>Bin Range Definition	O		8.3.26	Indicates the value ranges of bins for distribution type measurement
Granularity Period	M		8.3.8 Granularity Period	Collection interval of measurements
Cell Global ID	O		8.3.20 Cell Global ID	Points to a specific cell for generating measurements subscribed by the *Measurement Information List* IE. This IE is ignored when Report Style 255 is used, *Common Action Definition* IE is present and for the specific job, *Job Specific Action Definition* IE is absent and *Job Specific Cell Global ID* IE is present.

Range bound	Explanation
maxnoofMeasurementInfo	Maximum no. of measurement types that can be reported by a single report. Value is <65535>.
maxnoofConditionInfo	Maximum no. of conditions that can be subscribed for a single measurement type. Value is <32768>.

8.2.1.2.4 E2SM-KPM Action Definition Format 4

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Matching Condition		*1.. <maxnoofConditionInfoPerSub>*		The Matching Condition represents the Boolean expression, the logical connection to the next condition is AND if the *Logical OR* IE is not included
>Test Information	M		8.3.22 Test Condition Information	Provides test condition to filter matching UEs
>Logical OR	O		8.3.25	If included, logical connection to the next condition is "or".
Subscription Information	M		8.2.1.2.1 E2SM-KPM Action Definition Format 1	

Range bound	Explanation
maxnoofConditionInfoPerSub	Maximum no. of conditions that can be subscribed for a single subscription. Value is <32768>.

8.2.1.2.5 E2SM-KPM Action Definition Format 5

IE/Group Name	Presence	Range	IE type and reference	Semantics description
List of Subscribed UE IDs		*2.. <maxnoofUEIDPerSub>*		Points to a list of UEs of interest. This IE is ignored when Report Style 255 is used, *Common Action Definition* IE is present and for the specific job, *Job Specific Action Definition* IE is absent and *Job Specific List of Subscribed UE IDs* IE is present.
>UE ID	M		8.3.24	
Subscription Information	M		8.2.1.2.1 E2SM-KPM Action Definition Format 1	

Range bound	Explanation
maxnoofUEIDPerSub	Maximum no. of UE IDs that can be subscribed for a single subscription. Value is <65535>.

8.2.1.2.6 E2SM-KPM Action Definition Format 255

IE/Group Name	Presence	Range	IE type and reference	Semantics description
List of Multi Report Action Definitions		*1..<maxnoofJobs>*		
>Job ID	M		8.3.30	
>Job Specific Action Definition	O		8.2.1.2.7	Used to define the Action Definition for a specific job. If absent, then the *Common Action Definition* IE applies to this specific job
>Job Specific Cell Global ID	O		8.3.20	Used to modify the *Cell Global ID* IE within the *Common Action Definition* IE when the *Job Specific Action Definition* IE is not present for this specific job
>Job Specific UE ID	O		8.3.24	Used to modify the *UE ID* IE within the *Common Action Definition* IE when the *Job Specific Action Definition* IE is not present for this specific job
>Job Specific List of Subscribed UE IDs	O			Used to modify the *List of Subscribed UE IDs* IE within the *Common Action Definition* IE when *Job Specific Action Definition* IE is not present for this specific job
Reporting Approach	M		ENUMERATED (Single Integrated, Multiple Individually, ...)	Indicates if integrated report is to be transmitted as Single report or Multiple Reports with one per each job
Common Action Definition	O		8.2.1.2.7	Used to define a common Action Definition for one or more jobs. Shall be included if at least one job does not have the corresponding *Job Specific Action Definition* IE configured.

Range bound	Explanation
maxnoofJobs	Maximum no. of fundamental levels jobs that can be reported. Value is <65535>

8.2.1.2.7 E2SM-KPM Fundermental-level Action Definition Choice IE

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Fundermental-level Action Definition Format*	M			
>E2SM-KPM Action Definition Format 1			8.2.1.2.1	
>E2SM-KPM Action Definition Format 2			8.2.1.2.2	
>E2SM-KPM Action Definition Format 3			8.2.1.2.3	
>E2SM-KPM Action Definition Format 4			8.2.1.2.4	
>E2SM-KPM Action Definition Format 5			8.2.1.2.5	

8.2.1.3 RIC INDICATION HEADER IE

This information element is part of the RIC INDICATION message sent by the E2 Node to the Near-RT RIC and is required for REPORT action.

Direction: E2 Node → NEAR-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Indication Header Format*	M			
>E2SM-KPM Indication Header Format 1			8.2.1.3.1	

8.2.1.3.1 E2SM-KPM Indication Header Format 1

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Collection Start Time	M		8.3.12 Time Stamp	
File Format Version	O		PrintableString (SIZE (0..15), …)	
Sender Name	O		PrintableString (SIZE (0..400), …)	
Sender Type	O		PrintableString (SIZE (0..8), …)	
Vendor Name	O		PrintableString (SIZE (0..32), …)	
Job ID	O		8.3.30	Used only for the REPORT Service Style 255 when “Multiple Individually” reporting is configured, to indicate the associated job ID of the RIC INDICATION message.

8.2.1.4 RIC INDICATION MESSAGE IE

This information element is part of the RIC INDICATION message sent by the E2 Node to the Near-RT RIC and is required for REPORT action.

Direction: E2 Node → NEAR-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Indication Message Format*	M			
>E2SM-KPM Indication Message Format 1			8.2.1.4.1	
>E2SM-KPM Indication Message Format 2			8.2.1.4.2	
>E2SM-KPM Indication Message Format 3			8.2.1.4.3	
>E2SM-KPM Indication Message Format 255			8.2.1.4.4	Used only when "Single Integrated" reporting is configured for the REPORT Service Style 255.

8.2.1.4.1 E2SM-KPM Indication Message Format 1

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Measurements Data		*1..<maxnoofMeasurementRecord>*		Contains a set of Measurement Records, each collected at each Granularity Period.
>Measurements Record		*1..<maxnoofMeasurementValue>*		Contains measured values in same order as in the *Measurements Information List* IE if present, otherwise in the order defined in the subscription.
>>CHOICE *Measured Value*	M			
>>>Integer Value			INTEGER (0..4294967295)	
>>>Real Value			REAL	
>>>No Value			NULL	Indicates that no measurement value was generated.
>>>Not Satisfied			NULL	Indicates that the measured value did not satisfy the Matching Condition for Reporting, if configured.
>Incomplete Flag	O		ENUMERATED (true, …)	Indicates that the measurements record is not reliable.
Measurement Information List		*0..<maxnoofMeasurementInfo>*		
>CHOICE *Measurement Type*	M			
>>Measurement Name			8.3.9 Measurement Type Name	
>>Measurement ID			8.3.10 Measurement Type ID	
>List of Labels		*1..<maxnoofLabelInfo>*		
>>Label Information	M		8.3.11 Measurement Label	
>Matching Condition for Reporting		*0..<maxnoofConditionInfo>*		
>>Measured Value Reporting Condition	M		8.3.28	
>>Logical OR	O		8.3.25	
Granularity Period	O		8.3.8 Granularity Period	Collection interval of measurements

Range bound	Explanation
maxnoofMeasurementInfo	Maximum no. of measurement types that can be reported by a single report. Value is <65535>.
maxnoofConditionInfo	Maximum no. of conditions that can be subscribed for a single measurement type. Value is <32768>.
maxnoofLabelInfo	Maximum no. of measurements values that can be reported for a single measurement type. Value is <2147483647>.
maxnoofMeasurementRecord	Maximum no. of measurement records that can be reported by a single report. Value is <65535>.
maxnoofMeasurementValue	Maximum no. of measurement values that can be carried by a single measurement record. Value is <2147483647>.

8.2.1.4.2 E2SM-KPM Indication Message Format 2

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Measurements Data		*1..<maxnoofMeasurementRecord>*		Contains a set of Measurement Records, each collected at each Granularity Period.
>Measurements Record		*1..<maxnoofMeasurementValue>*		Contains measured values in same order as in the *Measurements Information Condition UE List* IE.
>>CHOICE *Measured Value*	M			
>>>Integer Value			INTEGER (0..4294967295)	
>>>Real Value			REAL	
>>>No Value			NULL	
>Incomplete Flag	O		ENUMERATED (true, …)	Indicates that the measurements record is not reliable.
Measurement Information Condition UE List		*1..<maxnoofMeasurementInfo>*		
>CHOICE *Measurement Type*	M			
>>Measurement Name			8.3.9 Measurement Type Name	
>>Measurement ID			8.3.10 Measurement Type ID	
>Matching Condition		*1..<maxnoofConditionInfo>*		
>>CHOICE *Condition Type*	M			
>>>Label Information			8.3.11 Measurement Label	
>>>Test Information			8.3.22 Test Condition Information	
>>Logical OR	O		8.3.25	
>List of matched UE IDs		*0..<maxnoofUEID>*		Indicates the UE ID(s) matched for the corresponding measurement type during the Reporting Period.
>>UE ID	M		8.3.24	
>Sequence of Matched UE IDs for Granularity Periods		*0..<maxnoofMeasurementRecord>*		Indicates the UE ID(s) matched for the corresponding measurement type, separately for each and every Granularity Period in chronological order. If included, the *List of matched UE IDs* IE shall be ignored if received.
>>CHOICE *Matched UE for Granularity Period*	M			
>>>*None*				
>>>>No UE matched for Granularity Period	M		ENUMERATED (true, …)	Indicates that none of UEs were matched for the corresponding measurement type for the corresponding Granularity Period.
>>>*One or more*				

>>>>List of UE IDs for Granularity Period		*1.. <maxnoofUEID>*		Indicates the UE ID(s) matched for the corresponding measurement type for the corresponding Granularity Period.
>>>>>UE ID	M		8.3.24	
Granularity Period	O		8.3.8 Granularity Period	Collection interval of measurements

Range bound	Explanation
maxnoofMeasurementInfo	Maximum no. of measurement types that can be reported by a single report. Value is <65535>.
maxnoofConditionInfo	Maximum no. of conditions that can be subscribed for a single measurement type. Value is <32768>.
maxnoofUEID	Maximum no. of UE IDs that can be reported for a single condition. Value is <65535>.
maxnoofMeasurementRecord	Maximum no. of measurement records that can be reported by a single report. Value is <65535>.
maxnoofMeasurementValue	Maximum no. of measurement values that can be carried by a single measurement record. Value is <2147483647>.

8.2.1.4.3 E2SM-KPM Indication Message Format 3

IE/Group Name	Presence	Range	IE type and reference	Semantics description
List of UE Measurement Reports		*1.. <maxnoofUEMeasReport>*		
>UE ID	M		8.3.24	
>Measurements Report	M		8.2.1.4.1 E2SM-KPM Indication Message Format 1	Contains Measurement Data for a UE for a Reporting Period.

Range bound	Explanation
maxnoofUEMeasReport	Maximum no. of UE Measurement Reports that can be reported. Value is <65535>.

8.2.1.4.4 E2SM-KPM Indication Message Format 255

IE/Group Name	Presence	Range	IE type and reference	Semantics description
List of Fundamental level Service style RIC Indication messages		*1.. <maxnoofJobs>*		
>Job ID	M		8.3.30	
>RIC Indication Message	M		8.2.1.4.5	Contains the job-specific report

Range bound	Explanation
maxnoofJobs	Maximum no. of fundamental levels jobs that can be reported. Value is <65535>.

8.2.1.4.5 E2SM-KPM Fundermental-level Indication Message Choice IE

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Fundermental-level Indication Message Format*	M			
>E2SM-KPM Indication Message Format 1			8.2.1.4.1	
>E2SM-KPM Indication Message Format 2			8.2.1.4.2	
>E2SM-KPM Indication Message Format 3			8.2.1.4.3	

8.2.1.5 RIC CALL PROCESS ID

Void.

8.2.1.6 RIC CONTROL HEADER IE

Void.

8.2.1.7 RIC CONTROL MESSAGE IE

Void.

8.2.1.8 RIC CONTROL OUTCOME IE

Void.

8.2.2 Messages for RIC Global Procedures

8.2.2.1 RAN Function Definition IE

This information element is part of the E2 SETUP REQUEST, and RIC SERVICE UPDATE message sent by the E2 Node to the Near-RT RIC and is used to provide all required information for the Near-RT RIC to determine how a given E2 Node has been configured to support a given RAN Function specific E2SM.

Direction: E2 Node → NEAR-RT RIC.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
RAN Function Name	M		8.3.2	
Sequence of Event Trigger styles		*0.. <maxnoofRICStyles>*		
>RIC Event Trigger Style Type	M		8.3.3	
>RIC Event Trigger Style Name	M		8.3.4	
>RIC Event Trigger Format Type	M		8.3.5	
Sequence of Report styles		*0.. <maxnoofRICStyles>*		To be used to describe supported fundamental level report styles
>RIC Report Style Type	M		8.3.3	
>RIC Report Style Name	M		8.3.4	
>RIC Report Action Format Type	M		8.3.5	
>Sequence of Measurement Info for Action		*1.. <maxnoofMeasurementInfo>*		
>>Measurement Type Name	M		8.3.9	
>>Measurement Type ID	O		8.3.10	
>>Bin Range Definition	O		8.3.26	Indicates the value ranges of bins for distribution type measurement
>RIC Indication Header Format Type	M		8.3.5	Indication header type used by Report style
>RIC Indication Message Format Type	M		8.3.5	Indication message type used by Report style
Sequence of Integrated level Report styles		*0.. <maxnoofRICStyles>*		To be used to describe supported integrated level report styles
>RIC Report Style Type	M		8.3.3	
>RIC Report Style Name	M		8.3.4	
>RIC Report Action Format Type	M		8.3.5	
>RIC Indication Header Format Type	M		8.3.5	Indication header type used by Report style
>RIC Indication Message Format Type	M		8.3.5	Indication message type used by Report style
>Sequence of Fundamental level report styles		*1.. <maxnoofRICStyles>*		
>>RIC Report Style Type	M		8.3.3	Supported features (formats, measurement info, etc.) defined in corresponding fundamental level Report Style
>Multiple Individual Reporting Support	O		ENUMERATED (true, …)	

Range bound	Explanation
maxnoofCells	Maximum no. of cells supported by an E2 Node component. The value is <16384>.
maxnoofRICstyle	Maximum no. of Style of Report, Insert, Control or Policy actions supported by RAN Function. The value is <63>.
maxnoofMeasurementInfo	Maximum no. of measurement types that can be reported by a single report. The value is <65535>.

8.3 Information Element definitions

8.3.1 General

When specifying information elements which are to be represented by bit strings, if not otherwise specifically stated in the semantics description of the concerned IE or elsewhere, the following principle applies with regards to the ordering of bits:

- The first bit (leftmost bit) contains the most significant bit (MSB);
- The last bit (rightmost bit) contains the least significant bit (LSB);
- When importing bit strings from other specifications, the first bit of the bit string contains the first bit of the concerned information.

8.3.2 RAN Function Name

This IE is defined in [12] clause 6.2.2.1.

8.3.3 RIC Style Type

This IE is defined in [12] clause 6.2.2.2.

8.3.4 RIC Style Name

This IE is defined in [12] clause 6.2.2.3.

8.3.5 RIC Format Type

This IE is defined in [12] clause 6.2.2.4.

8.3.6 Void

8.3.7 Void

8.3.8 Granularity Period

This IE defines the measurement collection interval within a reporting period.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Granularity Period	M		INTEGER (1..4294967295)	Measurement collection interval expressed in unit of 1 millisecond.

8.3.9 Measurement Type Name

This IE defines the name of a given measurement type.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Measurement Name	M		PrintableString(SIZE(1.. 150, ...))	One of the measurement names specified in TS 28.552 [4], TS 32.425 [8], or clause 7.10. The subcounters are represented by the Measurement Labels defined in 8.3.11.

8.3.10 Measurement Type ID

This IE defines the identifier of a given measurement type.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Measurement ID	M		INTEGER (1.. 65535, ...)	

8.3.11 Measurement Label

This IE defines values of necessary subcounters applicable to an associated measurement type.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
No Label	O		ENUMERATED (true, …)	Indicates the associated measurement type without any subcounter. If included, other IEs in 8.3.11 shall not be included in the same Measurement Label (and vice versa).
PLMN ID	O		8.3.15	Represents the PLMN subcounter
Slice ID	O		8.3.14	Represents the SNSSAI subcounter. OCTET STRING of length 1 octet shall be provided for matching the SST value only. OCTET STRING of length 4 octets shall be provided for matching the SST + SD value. OCTET STRING of length 4 octets with the last 3 octets as 0xFFFFFF shall be provided if a S-NSSAI without SD value has to be explicitly matched. See 3GPP TS 23.003 [20] clause 28.4.2.
5QI	O		8.3.17	Represents the 5QI subcounter
QFI	O		8.3.21	Represents the QFI subcounter
QCI	O		8.3.18	Represents the QCI subcounter
QCImax	O		8.3.18	Used only when the name of the associated measurement type ends with ‘_Filter’
QCImin	O		8.3.18	Used only when the name of the associated measurement type ends with ‘_Filter’

ARPmax	O		INTEGER (1.. 15, …)	Used only when the name of the associated measurement type ends with ‘_Filter’
ARPmin	O		INTEGER (1.. 15, …)	Used only when the name of the associated measurement type ends with ‘_Filter’
Bitrate Range	O		INTEGER (1.. 65535, ...)	Used only when the name of the associated measurement type ends with ‘_Filter’
Layer at MU-MIMO	O		INTEGER (1.. 65535, ...)	Represents the MIMO layer subcounter
Sum	O		ENUMERATED (true, …)	Sum is calculated as cumulative sum from the start of the measurement.
Distribution Bin X	O		INTEGER (1.. 65535, ...)	An index of Bin X. Only applicable to distribution type measurement information.
Distribution Bin Y	O		INTEGER (1.. 65535, ...)	An index of Bin Y. Only applicable to distribution type measurement information. This IE may be present only when Distribution Bin X is present.
Distribution Bin Z	O		INTEGER (1.. 65535, ...)	An index of Bin Z. Only applicable to distribution type measurement information. This IE may be present only when Distribution Bin X and Distribution Bin Y are present.
Precedent Label Override Indication	O		ENUMERATED (true, …)	Indicates that subcounters and their values of the precedent label applies in the same way except for the included subcounters. For included subcounters, new values shall apply.
Start End Indication	O		ENUMERATED (start, end, …)	Used to indicate a range of values. If “start” is used for a label, the subsequent label should include this IE with “end”. If included together with Distribution Bin type subcounter(s), it can be used to indicate a range of multi-dimensional values in the ascending order of numbers from Bin Z (if included), then from Bin Y (if included), then from Bin X (if included). In this case, information of a label with “start” should be identical to that of the subsequent label with “end”, except Distribution Bin type subcounter(s) used. Otherwise (included together with subcounter other than Distribution Bin type subcounter), it can be used to indicate one-dimensional range of values in the ascending order, and information of a label with “start” should be identical to that of the subsequent label with “end”, except only one subcounter.
Min	O		ENUMERATED (true, …)	Minimum of the measured values over a granularity period.
Max	O		ENUMERATED (true, …)	Maximum of the measured values over a granularity period.
Avg	O		ENUMERATED (true, …)	Average of the measured values over a granularity period.

SSB Index	O		INTEGER (1.. 65535, ...)	Represents the SSB subcounter
Non-GoB Beamforming Mode Index	O		INTEGER (1.. 65535, ...)	Represents the Non- Grid of Beams (Non-GoB) beamforming mode subcounter [21]. The index is used for Non-GoB beamforming optimization for 5G mMIMO deployments. Each BF mode implies a vendor-specific proprietary Non-GoB BF algorithm that are not standardized [i.1], for which each E2 Node, who supports the Non-GoB beamforming optimization feature, provides the number of different Non-GoB BF mode(s) supported by its scheduler indexed from 1 to n. The AI/ML model for Non-GoB beamforming optimization is trained by data and measurements related to each BF mode and/or MIMO mode, for which the trained AI/ML model, based on collected data, configures the E2 Node with the best inferred Non-GoB BF mode index to be used for each UE, where such configuration could be done separately for the case of Single User- and/or Multi-user MIMO [21].
MIMO Mode Index	O		INTEGER (1..2, …)	Represents the MIMO mode subcounter. Value = 1 means the SU (single-user) MIMO mode. Value 2 means the MU (multi-user) MIMO mode.
CGI	O		8.3.20	Represents the subcounter for a specific cell, e.g., a neighboring cell of the subscribed E2 Node.
Beam ID	O		8.3.29	Represents the beam subcounter, where the beam belongs to the subscribed E2 Node. An E2 Node level measurement can be obtained at beam-level using this subcounter.

8.3.12 Time Stamp

This IE contains UTC time information in picosecond-level. In this version of specification, the value is rounded in millisecond.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Time Stamp	M		OCTET STRING (SIZE(8))	Encoded in the same format as the 64-bit timestamp format as defined in clause 6 of IETF RFC 5905 [13].

8.3.13 Void

8.3.14 S-NSSAI

This IE is defined in [12] clause 6.2.3.12.

8.3.15 PLMN Identity

This IE is defined in [12] clause 6.2.3.1.

8.3.16 Void

8.3.17 5QI

This IE is defined in [12] clause 6.2.3.13.

8.3.18 QCI

This IE is defined in [12] clause 6.2.3.14.

8.3.19 Void

8.3.20 Cell Global ID

This IE is defined in [12] clause 6.2.2.5.

8.3.21 QFI

This IE is defined in [12] clause 6.2.3.15.

8.3.22 Test Condition Information

This IE defines a test condition for identifying UEs.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Test Condition Type*	M			
>GBR			ENUMERATED (true, …)	Identifies UEs with the GBR QoS flows or within the specified bitrate range. The definition of GBR QoS flow is in TS 23.501 [14].
>AMBR			ENUMERATED (true, …)	Identifies UEs with the Session-AMBR within the specified bitrate range. The definition of Session-AMBR is in TS 23.501 [14].
>IsStat			ENUMERATED (true, …)	This IE is not used in this version of specification.
>IsCatM			ENUMERATED (true, …)	This IE is not used in this version of specification.
>DL RSRP			ENUMERATED (true, …)	Identifies UEs with the latest reported DL RSRP measurement for this cell within the specified range. For EUTRAN, the definition of DL RSRP is in TS 36.214 [15]. For NR, the definition of DL RSRP is in TS 38.215 [16].
>DL RSRQ			ENUMERATED (true, …)	Identifies UEs with the latest reported DL RSRQ measurement for this cell within the specified range. For EUTRAN, the definition of DL RSRP is in TS 36.214 [15]. For NR, the definition of DL RSRQ is in TS 38.215 [16].
>UL RSRP			ENUMERATED (true, …)	Identifies UEs with the latest measured UL SRS-RSRP for this cell by the E2 Node within the specific range. The definition of UL SRS-RSRP is defined in TS 38.215 [16]. The mapping of measured quantity is described similarly to DL RSRP using the TS 38.133 [17] Table 10.1.6.1-1.
>CQI			ENUMERATED (true, …)	Identifies UEs with the latest reported wideband CQI for this cell in the Layer 1 within the specific range. The definition of wideband CQI is defined in TS 38.214 [18].
>5QI			ENUMERATED (true, …)	Identifies UEs with the 5QI of QoS flows within the specified range. The definition of 5QI is in TS 23.501 [14].
>QCI			ENUMERATED (true, …)	Identifies UEs with the QCI of Service Data Flows within the specified range. The definition of QCI is in TS 23.203 [19].
>S-NSSAI			ENUMERATED (true, …)	Identifies UEs with the S-NSSAI [12] within the specified range. OCTET STRING of length 1 octet shall be provided for matching the SST value only. OCTET STRING of length 4 octets shall be provided for matching the SST + SD value. OCTET STRING of length 4 octets with the last 3 octets as 0xFFFFFF shall be provided if a S-NSSAI without SD value has to be explicitly matched. See 3GPP TS 23.003 [20] clause 28.4.2.
Test Condition	O		ENUMERATED (equal, greaterthan, lessthan, contains, present, …)	
Test Condition Value	O		8.3.23	

8.3.23 Test Condition Value

This IE defines the target value for a particular Test Condition Type IE element.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Test Value*	M			
>INTEGER			INTEGER	
>ENUMERATED			INTEGER	
>BOOLEAN			BOOLEAN	
>BIT STRING			BIT STRING	
>OCTET STRING			OCTET STRING	
>PRINTABLE STRING			PrintableString	
>REAL			REAL	

8.3.24 UE ID

This IE is defined in [12] clause 6.2.2.6.

8.3.25 Logical OR

This IE indicates a logical “or” connection of the current condition to the next condition in a given sequence of conditions.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Logical OR	M		ENUMERATED (true, …)	If set to “true”, logical connection to the next condition is “or”.

8.3.26 Bin Range Definition

This IE defines the value range of bins for distribution type measurements.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
List of Bins for Distribution Bin X		*1.. <maxnoofBin>*		
>Bin Index	M		INTEGER (1.. 65535, ..)	Index of a bin to be used when subscribed
>Start Value	M		8.3.27	
>End Value	M		8.3.27	
List of Bins for Distribution Bin Y		*0.. <maxnoofBin>*		Shall not be included for a distribution measurement type that doesn't use Distribution Bin Y.
>Bin Index	M		INTEGER (1.. 65535, ..)	Index of a bin to be used when subscribed
>Start Value	M		8.3.27	
>End Value	M		8.3.27	
List of Bins for Distribution Bin Z		*0.. <maxnoofBin>*		Shall not be included for a distribution measurement type that doesn't use Distribution Bin Z.
>Bin Index	M		INTEGER (1.. 65535, ..)	Index of a bin to be used when subscribed
>Start Value	M		8.3.27	
>End Value	M		8.3.27	

Range bound	Explanation
maxnoofBin	Maximum no. of bins that can be defined for a distribution type measurement. Value is <65535>.

8.3.27 Bin Range Value

This IE defines either the start or end value of a bin for distribution type measurements.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Bin Range Value*	M			
>INTEGER			INTEGER	
>REAL			REAL	

8.3.28 Measured Value Reporting Condition

This IE defines a test condition for reporting measured values when satisfied.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Test Condition	M		ENUMERATED (equal, greaterthan, lessthan, contains, present, …)	
Test Condition Value	M		8.3.23	

8.3.29 Beam ID

This IE is defined in [12] clause 6.2.2.16.

8.3.30 Job ID

This IE indicates the identifier of individual reports when using integrated level services.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Job ID	M		INTEGER (1.. *maxnoofJobs*)	

8.3.31 Event Formula

This IE defines the logical structure of an event condition that determines whether an event trigger is satisfied. This IE consists of an expression on the left-hand side, a logical operator in the middle (defining the type of comparison), and an expression on the right-hand side. The event is considered triggered when the comparison evaluates as true.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Event Left Expression	M		8.3.32 Event Expression	Left-hand side of the event formula
Event Logical Operator	M		ENUMERATED (equal, greaterthan, lessthan, …)	Logical operator for comparison
Event Right Expression	M		8.3.32 Event Expression	Right-hand side of the event formula

8.3.32 Event Expression

This IE defines an event expression with measurement-based or value-based parameters.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Sequence of Parameters		*1.. <maxnoofEventParameters>*		
>**CHOICE** *Parameter Type*	M			Specifies whether a parameter is a measurement or a value
>>***Measurement***				
>>>Measurement Definition	M		8.3.33 Event Measurement	
>>>Window Definition	O		8.3.34 Window Expression	Applies a window to aggregate multiple samples
>>***Value***				
>>>Test Value	M		8.3.23 Test Condition Value	
>Next Mathematical Operator	O		ENUMERATED (plus, minus, product, divide-by, …)	Operators to combine parameters
Aggregation Function	O		8.3.35 Aggregation Function Type	

Range bound	Explanation
maxnoofEventParameters	Maximum number of event parameters. Value is <65535>.

8.3.33 Event Measurement

This IE identifies the specific measurement(s), whose measured values are used in the event expression evaluation.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
CHOICE *Measurement Type*	M			
>Measurement Name			8.3.9 Measurement Type Name	
>Measurement ID			8.3.10 Measurement Type ID	
List of Labels		*1..<maxnoofLabelInfo>*		
>Label Information	M		8.3.11 Measurement Label	
Cell Global ID	O		8.3.20 Cell Global ID	

Range bound	Explanation
maxnoofLabelInfo	Maximum no. of measurements values that can be reported for a single measurement type. Value is <2147483647>.

8.3.34 Window Expression

This IE defines windowing parameters for aggregating multiple measurement samples.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Window Size	M		INTEGER (1..100)	Number of samples to be considered
Aggregation Function	M		8.3.35 Aggregation Function Type	

8.3.35 Aggregation Function Type

This IE defines the mathematical operations to be applied for multiple values.

IE/Group Name	Presence	Range	IE type and reference	Semantics description
Aggregation Function Type	M		ENUMERATED (min, max, avg, sum, …)	

8.4 Information Element Abstract Syntax (with ASN.1)

8.4.1 General

E2SM-KPM ASN.1 definition conforms to ITU-T Rec. X.680 [5] and ITU-T Rec. X.681 [6].

Sub clause 8.4.2 presents the Abstract Syntax of the E2SM information elements to be carried within the E2AP [3] protocol messages with ASN.1. In case there is contradiction between the ASN.1 definition in this sub clause and the tabular format in sub clause 8.2 and 8.3, the ASN.1 shall take precedence, except for the definition of conditions for the presence of conditional elements, in which the tabular format shall take precedence.

If an E2SM information element carried as an OCTET STRING in an E2AP [3] message that is not constructed as defined above is received, this shall be considered as Abstract Syntax Error, and the message shall be handled as defined for Abstract Syntax Error in clause 9.

8.4.2 Information Element definitions

```
-- ASN1START
-- **************************************************************
-- E2SM-KPM Information Element Definitions
-- **************************************************************

E2SM-KPM-IEs {
iso(1) identified-organization(3) dod(6) internet(1) private(4) enterprise(1) 53148 e2(1) version2(2) e2sm(2) e2sm-KPMMON-IEs (2)}

DEFINITIONS AUTOMATIC TAGS ::=

BEGIN

-- **************************************************************
--   IEs
-- **************************************************************

IMPORTS
   CGI,
   FiveQI,
   PLMNIdentity,
   QCI,
   QosFlowIdentifier,
   RANfunction-Name,
   RIC-Format-Type,
   RIC-Style-Name,
   RIC-Style-Type,
   S-NSSAI,
   UEID,
   Beam-ID
```



```
FROM E2SM-COMMON-IEs;

TimeStamp ::= OCTET STRING (SIZE(8))

AggregationFunctionType ::= ENUMERATED {min, max, avg, sum, ...}

BinIndex ::= INTEGER (1.. 65535, ...)

BinRangeValue ::= CHOICE {
   valueInt          INTEGER,
   valueReal         REAL,
   ...
}

EventEvaluationPeriod ::= INTEGER (1.. 4294967295)

EventExpression ::= SEQUENCE {
   paramList         EventParameterList,
   aggreFunc         AggregationFunctionType          OPTIONAL,
   ...
}

EventFormula ::= SEQUENCE {
   leftExpr          EventExpression,
   logicOp           EventLogicalOperator,
   rightExpr         EventExpression,
   ...
}

EventLogicalOperator ::= ENUMERATED {equal, greaterthan, lessthan, ...}

EventMeasurementDefinition ::= SEQUENCE {
   measType                   MeasurementType,
   labelInfoList              LabelInfoList,
   cellGlobalID               CGI                     OPTIONAL,
   ...
}

EventParameterMeasurement ::= SEQUENCE {
   measDef           EventMeasurementDefinition,
   windowFunc        EventWindowDefinition            OPTIONAL,
   ...
}

EventParameterValue ::= SEQUENCE {
   value             TestCond-Value                   OPTIONAL,
   ...
}
```



```
EventWindowDefinition ::= SEQUENCE {
   windowSize        WindowSize,
   aggreFunc         AggregationFunctionType,
   ...
}

JobID ::= INTEGER (1.. 65535)


GranularityPeriod ::= INTEGER (1.. 4294967295)

LogicalOR ::= ENUMERATED {true, ...}

MeasurementType ::= CHOICE {
   measName          MeasurementTypeName,
   measID            MeasurementTypeID,
   ...
}

MeasurementTypeName ::= PrintableString(SIZE(1.. 150, ...))

MeasurementTypeID ::= INTEGER (1.. 65536, ...)

MeasurementLabel ::= SEQUENCE {
   noLabel              ENUMERATED {true, ...}           OPTIONAL,
   plmnID               PLMNIdentity                     OPTIONAL,
   sliceID              S-NSSAI                          OPTIONAL,
   fiveQI               FiveQI                           OPTIONAL,
   qFI                  QosFlowIdentifier                OPTIONAL,
   qCI                  QCI                              OPTIONAL,
   qCImax               QCI                              OPTIONAL,
   qCImin               QCI                              OPTIONAL,
   aRPmax               INTEGER (1.. 15, ...)            OPTIONAL,
   aRPmin               INTEGER (1.. 15, ...)            OPTIONAL,
   bitrateRange         INTEGER (1.. 65535, ...)         OPTIONAL,
   layerMU-MIMO         INTEGER (1.. 65535, ...)         OPTIONAL,
   sUM                  ENUMERATED {true, ...}           OPTIONAL,
   distBinX             INTEGER (1.. 65535, ...)         OPTIONAL,
   distBinY             INTEGER (1.. 65535, ...)         OPTIONAL,
   distBinZ             INTEGER (1.. 65535, ...)         OPTIONAL,
   preLabelOverride     ENUMERATED {true, ...}           OPTIONAL,
   startEndInd          ENUMERATED {start, end, ...}     OPTIONAL,
   min                  ENUMERATED {true, ...}           OPTIONAL,
   max                  ENUMERATED {true, ...}           OPTIONAL,
   avg                  ENUMERATED {true, ...}           OPTIONAL,
   ...,
   ssbIndex             INTEGER (1.. 65535, ...)         OPTIONAL,
   nonGoB-BFmode-Index  INTEGER (1.. 65535, ...)         OPTIONAL,
   mIMO-mode-Index      INTEGER (1.. 2, ...)             OPTIONAL,
```

```
   cellGlobalID        CGI                              OPTIONAL,
   beamID              Beam-ID                          OPTIONAL
}

TestCondInfo ::= SEQUENCE{
   testType          TestCond-Type,
   testExpr          TestCond-Expression                OPTIONAL,
   testValue         TestCond-Value                     OPTIONAL,
   ...
}

NextMathOperator ::= ENUMERATED {plus, minus, product, divide-by, ...}

TestCond-Type ::= CHOICE{
   gBR               ENUMERATED {true, ...},
   aMBR              ENUMERATED {true, ...},
   isStat            ENUMERATED {true, ...},
   isCatM            ENUMERATED {true, ...},
   rSRP              ENUMERATED {true, ...},
   rSRQ              ENUMERATED {true, ...},
   ...,
   ul-rSRP           ENUMERATED {true, ...},
   cQI               ENUMERATED {true, ...},
   fiveQI            ENUMERATED {true, ...},
   qCI               ENUMERATED {true, ...},
   sNSSAI            ENUMERATED {true, ...}
}

TestCond-Expression ::= ENUMERATED {
   equal,
   greaterthan,
   lessthan,
   contains,
   present,
   ...
}

TestCond-Value ::= CHOICE{
   valueInt          INTEGER,
   valueEnum         INTEGER,
   valueBool         BOOLEAN,
   valueBitS         BIT STRING,
   valueOctS         OCTET STRING,
   valuePrtS         PrintableString,
   ...,
   valueReal         REAL
}

MultiReportApproach ::= ENUMERATED {
   single,
```



```
   multiple,
   ...
}

WindowSize ::= INTEGER (1.. 100)

-- **************************************************************
--   Lists
-- **************************************************************

maxnoofCells                INTEGER ::= 16384
maxnoofRICStyles            INTEGER ::= 63
maxnoofMeasurementInfo      INTEGER ::= 65535
maxnoofLabelInfo            INTEGER ::= 2147483647
maxnoofMeasurementRecord    INTEGER ::= 65535
maxnoofMeasurementValue     INTEGER ::= 2147483647
maxnoofConditionInfo        INTEGER ::= 32768
maxnoofUEID                 INTEGER ::= 65535
maxnoofConditionInfoPerSub INTEGER ::= 32768
maxnoofUEIDPerSub           INTEGER ::= 65535
maxnoofUEMeasReport         INTEGER ::= 65535
maxnoofBin                  INTEGER ::= 65535
maxnoofJobs                 INTEGER ::= 65535
maxnoofEventParameters      INTEGER ::= 65535
maxnoofEventTrigger         INTEGER ::= 65535


BinRangeDefinition ::= SEQUENCE {
   binRangeListX     BinRangeList,
   binRangeListY     BinRangeList         OPTIONAL -- This IE shall not be present for a distribution measurement type that doesn't use Distribution Bin Y --,
   binRangeListZ     BinRangeList         OPTIONAL -- This IE shall not be present for a distribution measurement type that doesn't use Distribution Bin Z --,
   ...
}

BinRangeList ::= SEQUENCE (SIZE(1..maxnoofBin)) OF BinRangeItem

BinRangeItem ::= SEQUENCE {
   binIndex          BinIndex,
   startValue        BinRangeValue,
   endValue          BinRangeValue,
   ...
}

DistMeasurementBinRangeList ::= SEQUENCE (SIZE(1..maxnoofMeasurementInfo)) OF DistMeasurementBinRangeItem

DistMeasurementBinRangeItem ::= SEQUENCE {
   measType          MeasurementType,
   binRangeDef       BinRangeDefinition,
   ...
}
```

```
EventParameterList ::= SEQUENCE (SIZE(1..maxnoofEventParameters)) OF EventParameterItem

EventParameterItem ::= SEQUENCE {
   paramType              CHOICE {
      measurement            EventParameterMeasurement,
      value                  EventParameterValue,
      ...
   },
   nextMathOp             NextMathOperator          OPTIONAL,
   ...
}

EventTrigerList ::= SEQUENCE (SIZE(1..maxnoofEventTrigger)) OF EventTriggerItem

EventTriggerItem ::= SEQUENCE {
   eventCond           EventFormula,
   eventEvalPeriod     EventEvaluationPeriod      OPTIONAL,
   logicalOR           LogicalOR                  OPTIONAL,
   ...
}

MeasurementInfoList ::= SEQUENCE (SIZE(1..maxnoofMeasurementInfo)) OF MeasurementInfoItem

MeasurementInfoItem ::= SEQUENCE {
   measType               MeasurementType,
   labelInfoList          LabelInfoList,
   ...,
   matchCondReportList    MatchCondReportList          OPTIONAL

}

LabelInfoList ::= SEQUENCE (SIZE(1..maxnoofLabelInfo)) OF LabelInfoItem

LabelInfoItem ::= SEQUENCE {
   measLabel         MeasurementLabel,
   ...
}

MatchCondReportList ::= SEQUENCE (SIZE(1..maxnoofConditionInfo)) OF MatchCondReportItem

MatchCondReportItem ::= SEQUENCE {
   measValueReportCond       MeasValueReportCond,
   logicalOR                 LogicalOR                  OPTIONAL,
   ...
}

MeasValueReportCond ::= SEQUENCE {
   testExpr          MeasValueTestCond-Expression,
```

```
   testValue         TestCond-Value,
   ...
}

MeasValueTestCond-Expression ::= ENUMERATED {
   equal,
   greaterthan,
   lessthan,
   contains,
   present,
   ...
}

MeasurementData ::= SEQUENCE (SIZE(1..maxnoofMeasurementRecord)) OF MeasurementDataItem

MeasurementDataItem ::= SEQUENCE {
   measRecord          MeasurementRecord,
   incompleteFlag      ENUMERATED {true, ...}              OPTIONAL,
   ...
}

MeasurementRecord ::= SEQUENCE (SIZE(1..maxnoofMeasurementValue)) OF MeasurementRecordItem

MeasurementRecordItem ::= CHOICE {
   integer           INTEGER (0.. 4294967295),
   real              REAL,
   noValue           NULL,
   ...,
   notSatisfied      NULL

}

MeasurementInfo-Action-List ::= SEQUENCE (SIZE(1..maxnoofMeasurementInfo)) OF MeasurementInfo-Action-Item

MeasurementInfo-Action-Item ::= SEQUENCE {
   measName          MeasurementTypeName,
   measID            MeasurementTypeID          OPTIONAL,
   ...,
   binRangeDef       BinRangeDefinition         OPTIONAL
}

MeasurementCondList ::= SEQUENCE (SIZE(1..maxnoofMeasurementInfo)) OF MeasurementCondItem

MeasurementCondItem ::= SEQUENCE {
   measType          MeasurementType,
   matchingCond      MatchingCondList,
   ...,
   binRangeDef       BinRangeDefinition         OPTIONAL
}
```



```
MeasurementCondUEidList ::= SEQUENCE (SIZE(1..maxnoofMeasurementInfo)) OF MeasurementCondUEidItem

MeasurementCondUEidItem ::= SEQUENCE {
   measType             MeasurementType,
   matchingCond         MatchingCondList,
   matchingUEidList     MatchingUEidList        OPTIONAL,
   ...,
   matchingUEidPerGP    MatchingUEidPerGP       OPTIONAL
}

MatchingCondList ::= SEQUENCE (SIZE(1..maxnoofConditionInfo)) OF MatchingCondItem

MatchingCondItem ::= SEQUENCE {
   matchingCondChoice   MatchingCondItem-Choice,
   logicalOR            LogicalOR                  OPTIONAL,
   ...
}

MatchingCondItem-Choice ::= CHOICE{
   measLabel         MeasurementLabel,
   testCondInfo      TestCondInfo,
   ...
}

MatchingUEidList ::= SEQUENCE (SIZE(1..maxnoofUEID)) OF MatchingUEidItem

MatchingUEidItem ::= SEQUENCE{
   ueID              UEID,
   ...
}

MatchingUEidPerGP ::= SEQUENCE (SIZE(1..maxnoofMeasurementRecord)) OF MatchingUEidPerGP-Item

MatchingUEidPerGP-Item ::= SEQUENCE{
   matchedPerGP            CHOICE{
      noUEmatched             ENUMERATED {true, ...},
      oneOrMoreUEmatched      MatchingUEidList-PerGP,
      ...
   },
   ...
}

MatchingUEidList-PerGP ::= SEQUENCE (SIZE(1..maxnoofUEID)) OF MatchingUEidItem-PerGP

MatchingUEidItem-PerGP ::= SEQUENCE{
   ueID              UEID,
   ...
}
```



```
MatchingUeCondPerSubList ::= SEQUENCE (SIZE(1..maxnoofConditionInfoPerSub)) OF MatchingUeCondPerSubItem

MatchingUeCondPerSubItem ::= SEQUENCE{
   testCondInfo       TestCondInfo,
   ...,
   logicalOR          LogicalOR            OPTIONAL
}

MatchingUEidPerSubList ::= SEQUENCE (SIZE(2..maxnoofUEIDPerSub)) OF MatchingUEidPerSubItem

MatchingUEidPerSubItem ::= SEQUENCE{
   ueID               UEID,
   ...
}

MultiReportActionDefinitionList ::= SEQUENCE (SIZE(1..maxnoofJobs)) OF MultiReportActionDefinitionItem

MultiReportActionDefinitionItem ::=SEQUENCE {
   jobID              JobID,
   jobActionDefinition     E2SM-KPM-ActionDefinition-Fundamental       OPTIONAL,
   jobCellGlobalID         CGI                              OPTIONAL,
   jobUEID              UEID                               OPTIONAL,
   jobMatchingUEidList     MatchingUEidPerSubList           OPTIONAL,
   ...
}

MultiReportIndicationMessageList ::= SEQUENCE (SIZE(1..maxnoofJobs)) OF MultiReportIndicationMessageItem


MultiReportIndicationMessageItem ::=SEQUENCE {
   jobID              JobID,
   indicationMessage    E2SM-KPM-IndicationMessage-Fundamental,
   ...
}


UEMeasurementReportList ::= SEQUENCE (SIZE(1..maxnoofUEMeasReport)) OF UEMeasurementReportItem

UEMeasurementReportItem ::= SEQUENCE{
   ueID           UEID,
   measReport     E2SM-KPM-IndicationMessage-Format1,
   ...
}



-- **************************************************************
-- E2SM-KPM Service Model IEs
-- **************************************************************
```



```
-- **************************************************************
--   Event Trigger Definition OCTET STRING contents
-- **************************************************************

E2SM-KPM-EventTriggerDefinition ::= SEQUENCE{
   eventDefinition-formats       CHOICE{
      eventDefinition-Format1       E2SM-KPM-EventTriggerDefinition-Format1,
      ...,
      eventDefinition-Format2       E2SM-KPM-EventTriggerDefinition-Format2
   },
   ...
}

E2SM-KPM-EventTriggerDefinition-Format1 ::= SEQUENCE{
   reportingPeriod               INTEGER (1.. 4294967295),
   ...
}

E2SM-KPM-EventTriggerDefinition-Format2 ::= SEQUENCE {
   eventTriggerList              EventTrigerList,
   ...
}

-- **************************************************************
--   Action Definition OCTET STRING contents
-- **************************************************************

E2SM-KPM-ActionDefinition ::= SEQUENCE{
   ric-Style-Type             RIC-Style-Type,
   actionDefinition-formats   CHOICE{
      actionDefinition-Format1      E2SM-KPM-ActionDefinition-Format1,
      actionDefinition-Format2      E2SM-KPM-ActionDefinition-Format2,
      actionDefinition-Format3      E2SM-KPM-ActionDefinition-Format3,
      ...,
      actionDefinition-Format4      E2SM-KPM-ActionDefinition-Format4,
      actionDefinition-Format5      E2SM-KPM-ActionDefinition-Format5,
      actionDefinition-Format255    E2SM-KPM-ActionDefinition-Format255
   },
   ...
}

E2SM-KPM-ActionDefinition-Fundamental ::= SEQUENCE{
   actionDefinitionFundamental-formats    CHOICE{
      actionDefinition-Format1      E2SM-KPM-ActionDefinition-Format1,
      actionDefinition-Format2      E2SM-KPM-ActionDefinition-Format2,
      actionDefinition-Format3      E2SM-KPM-ActionDefinition-Format3,
      actionDefinition-Format4      E2SM-KPM-ActionDefinition-Format4,
      actionDefinition-Format5      E2SM-KPM-ActionDefinition-Format5,
      ...
```



```
   },
   ...
}

E2SM-KPM-ActionDefinition-Format1 ::= SEQUENCE {
   measInfoList               MeasurementInfoList,
   granulPeriod               GranularityPeriod,
   cellGlobalID               CGI                     OPTIONAL,
   ...,
   distMeasBinRangeInfo       DistMeasurementBinRangeList      OPTIONAL
}

E2SM-KPM-ActionDefinition-Format2 ::= SEQUENCE {
   ueID                       UEID,
   subscriptInfo              E2SM-KPM-ActionDefinition-Format1,
   ...
}

E2SM-KPM-ActionDefinition-Format3 ::= SEQUENCE {
   measCondList               MeasurementCondList,
   granulPeriod               GranularityPeriod,
   cellGlobalID               CGI                     OPTIONAL,
   ...
}

E2SM-KPM-ActionDefinition-Format4 ::= SEQUENCE {
   matchingUeCondList         MatchingUeCondPerSubList,
   subscriptionInfo           E2SM-KPM-ActionDefinition-Format1,
   ...
}

E2SM-KPM-ActionDefinition-Format5 ::= SEQUENCE {
   matchingUEidList           MatchingUEidPerSubList,
   subscriptionInfo           E2SM-KPM-ActionDefinition-Format1,
   ...
}

E2SM-KPM-ActionDefinition-Format255 ::= SEQUENCE {
   multiReportActionDefinitionList     MultiReportActionDefinitionList,
   multiReportApproach              MultiReportApproach,
   commonActionDefinition           E2SM-KPM-ActionDefinition-Fundamental     OPTIONAL,
   ...
}

-- **************************************************************
--   Indication Header OCTET STRING contents
-- **************************************************************

E2SM-KPM-IndicationHeader ::= SEQUENCE{
   indicationHeader-formats      CHOICE{
```



```
      indicationHeader-Format1      E2SM-KPM-IndicationHeader-Format1,
      ...
   },
   ...
}

E2SM-KPM-IndicationHeader-Format1 ::= SEQUENCE{
   colletStartTime            TimeStamp,
   fileFormatversion          PrintableString (SIZE (0..15), ...)    OPTIONAL,
   senderName                 PrintableString (SIZE (0..400), ...)   OPTIONAL,
   senderType                 PrintableString (SIZE (0..8), ...)     OPTIONAL,
   vendorName                 PrintableString (SIZE (0..32), ...)    OPTIONAL,
   ...,
   jobID                      JobID                                  OPTIONAL
}

-- **************************************************************
--   Indication Message OCTET STRING contents
-- **************************************************************

E2SM-KPM-IndicationMessage ::= SEQUENCE{
   indicationMessage-formats     CHOICE{
      indicationMessage-Format1     E2SM-KPM-IndicationMessage-Format1,
      indicationMessage-Format2     E2SM-KPM-IndicationMessage-Format2,
      ...,
      indicationMessage-Format3     E2SM-KPM-IndicationMessage-Format3,
      indicationMessage-Format255      E2SM-KPM-IndicationMessage-Format255
   },
   ...
}

E2SM-KPM-IndicationMessage-Fundamental ::= SEQUENCE{
   indicationMessageFundamental-formats   CHOICE{
      indicationMessage-Format1     E2SM-KPM-IndicationMessage-Format1,
      indicationMessage-Format2     E2SM-KPM-IndicationMessage-Format2,
      indicationMessage-Format3     E2SM-KPM-IndicationMessage-Format3,
      ...
   },
   ...
}


E2SM-KPM-IndicationMessage-Format1 ::= SEQUENCE {
   measData                   MeasurementData,
   measInfoList               MeasurementInfoList                 OPTIONAL,
   granulPeriod               GranularityPeriod                   OPTIONAL,
   ...
}
```



```
E2SM-KPM-IndicationMessage-Format2 ::= SEQUENCE {
   measData                    MeasurementData,
   measCondUEidList            MeasurementCondUEidList,
   granulPeriod                GranularityPeriod                   OPTIONAL,
   ...
}

E2SM-KPM-IndicationMessage-Format3 ::= SEQUENCE {
   ueMeasReportList            UEMeasurementReportList,
   ...
}

E2SM-KPM-IndicationMessage-Format255 ::= SEQUENCE {
   multiReportIndicationMessageList    MultiReportIndicationMessageList,
   ...
}


-- ***************************************************************
--   RAN Function Definition OCTET STRING contents
-- ***************************************************************

E2SM-KPM-RANfunction-Description ::= SEQUENCE{
   ranFunction-Name               RANfunction-Name,
   ric-EventTriggerStyle-List     SEQUENCE (SIZE(1..maxnoofRICStyles)) OF RIC-EventTriggerStyle-Item     OPTIONAL,
   ric-ReportStyle-List           SEQUENCE (SIZE(1..maxnoofRICStyles)) OF RIC-ReportStyle-Item           OPTIONAL,
   ...,
   ric-IntegratedReportStyle-List      SEQUENCE (SIZE(1..maxnoofRICStyles)) OF RIC-IntegratedReportStyle-Item    OPTIONAL
}

RIC-EventTriggerStyle-Item ::= SEQUENCE{
   ric-EventTriggerStyle-Type       RIC-Style-Type,
   ric-EventTriggerStyle-Name       RIC-Style-Name,
   ric-EventTriggerFormat-Type      RIC-Format-Type,
   ...
}

RIC-ReportStyle-Item ::= SEQUENCE{
   ric-ReportStyle-Type                RIC-Style-Type,
   ric-ReportStyle-Name                RIC-Style-Name,
   ric-ActionFormat-Type               RIC-Format-Type,
   measInfo-Action-List                MeasurementInfo-Action-List,
   ric-IndicationHeaderFormat-Type     RIC-Format-Type,
   ric-IndicationMessageFormat-Type    RIC-Format-Type,
   ...
}

RIC-IntegratedReportStyle-Item ::= SEQUENCE{
   ric-ReportStyle-Type                   RIC-Style-Type,
   ric-ReportStyle-Name                   RIC-Style-Name,
```



```
   ric-ActionFormat-Type                    RIC-Format-Type,
   ric-IndicationHeaderFormat-Type          RIC-Format-Type,
   ric-IndicationMessageFormat-Type         RIC-Format-Type,
   ric-FundamentalLevelReportStyle-List     SEQUENCE (SIZE(1..maxnoofRICStyles)) OF RIC-FundamentalLevelReportStyle-Item,
   ric-MultipleIndividualReportingSupport   ENUMERATED{true,...}        OPTIONAL,
   ...
}

RIC-FundamentalLevelReportStyle-Item ::= SEQUENCE{
   ric-ReportStyle-Type                     RIC-Style-Type,
   ...
}

END

-- ASN1STOP
```

9 Handling of Unknown, Unforeseen and Erroneous Protocol Data

clause 10 of TS 36.413 [13] is applicable for the purposes of the present document.

Annex A (normative or informative):
Further information on RAN Function Network KPM Monitor

A.1 Background Information

The RAN function "Key Performance Measurement" is used to provide RIC Service exposure of the performance measurement logical function of the E2 Nodes. Based on the O-RAN deployment architecture, available measurements could be different. Figure A.1-1 shows the target deployment architecture for E2SM-KPM.

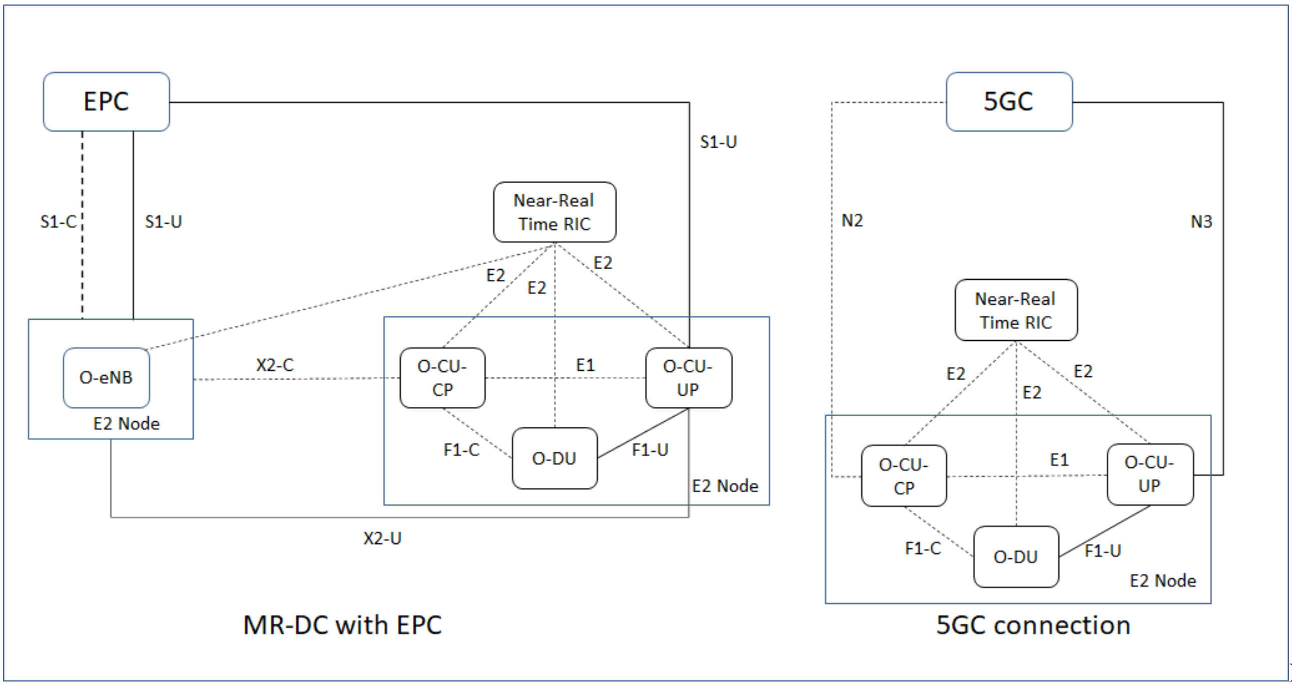

Figure A.1-1 E2SM-KPM Architecture

For each logical function the E2 Node shall use the RAN Function Definition IE to declare the list of available measurements and a set of supported RIC Services (**REPORT**).

Annex (informative): Change history/Change request (history)

Date	Revision	Description
2020.01.29	v01.00	Initial version
2020.12.16	02.00.00	Adopt INTEL.AO's CR-0001 and CR-0002 for E2SM-KPM with cleaning up old texts and ASN.1 in v01.00
2021.02.24	02.00.01	Adopt ATT.AO's CR-0001 for UE-level measurements subscription and retrieval
2021.03.03	02.00.02	Adopt CSP.AO's CR-0001 for Incomplete Flag
2021.03.30	02.00.03	Adopt INTEL's CR-0003 for clean-up
2021.06.09	02.00.04	Adopt (1) INT's CR-0006; (2) INT's CR-0008; (3) RSYS's CR-0004
2021.07.09	02.00.05	Adopt (1) INT's CR-0009; (2) NEU.AO's CR-0001
2021.08.10	02.00	New features: UE-level measurements subscription and retrieval, Incomplete Flag, clean up from v01.
2021.10.13	02.01.00	Adopt (1) INTEL.AO's CR-0011; (2) RSYS.AO's CR-0002
2021.10.27	02.01.01	Adopt KDDI's CR-0001.
2021.11.22	02.01.02	Editorial Updates based on review comments during WG3 approval process
2022.02.07	02.01	New features: Add Cell Global ID description new Report Service Styles 4 and 5 introduced for flexible reporting of UE level measurements, enhancing conversioin rule from measuments parameter defined in 3GPP and O-RAN, and added new O-RAN specific parameters
2022.03.23	02.02.00	Adopt (1) TIM.AO's CR-0003; (2) CMCC.AO's CR-0001; (3) CMCC.AO's CR-0002
2022.04.14	02.02	New features: Add new parameters for better supporting UE grouping with Format 3, add new test condition types 5QI and QCI for Report Service Style Type 3 and 4, and add S-NSSAI for new test condition.
2022.05.11	02.02.01	Adopt (1) INT's CR-0015; (2) INT's CR-0016; (3) INT's CR-0017
2022.07.20	02.02.02	Adopt INT's CR-0022
2022.07.24	02.02.03	Aligned to new template
2022.08.09	02.03	New features: Add new Report Service Styles 4 and 5 introduced for flexible reporting of UE level measurements, Enhancing conversion rule from measurements parameter defined in 3GPP and O-RAN adding new O-RAN specific parameters clarifying CGI handling in Action Definition Formats, UL-RSRP and CQI for signal quality based UE grouping in Format 3, extend new test condition types for Report Service Style Type 3 and 4, adding Slice ID measurement label, Adding "Logical OR" into UE filtering conditions, Indication Message Format 2 for REPORT Style 3, bin range definition of distribution type measurements, Support mMIMO non-GoB measurements
2022.11.09	02.03.01	Adopt (1) INT's CR-0024; (2) INT's CR-0027
2022.11.20	02.03.02	Editorial changes reflecting comments received during WG3 approval process
2022.11.25	03.00	New features: Correction for E2SM-KPM Timestamp, Correction for mMIMO Non-GoB BF optimization
2023.06.30	03.00.01	Adopt (1) NOK.AO's CR-0002; (2) MAV.AO's CR-0013
2023.07.26	03.00.02	Editorial changes reflecting comments received during WG3 approval process
2023.07.26	03.00.03	Further Editorial changes reflecting comments received during WG3 approval process
2023.07.29	04.00	New features: Add support mMIMO bMRO measurements and reporting, DL total number of scheduled time slots
2024.03.05	04.00.01	Adopt (1) SAM.AO's CR-0002; (2) NOK's CR-0003; (3) SAM's CR-0004 (4)SAM's CR-0006 (5) NOK.AO CR-0004 (6) SAM's CR-0007
2024.03.22	04.00.02	Adopt SAM's CR-0003
2024.03.27	04.00.03	Editorial changes reflecting comments received during WG3 approval process
2024.04.08	05.00	New feature: Add Multi-Cell support
2024.11.21	05.00.01	Adopt (1) SAM's CR-0006; (2) SAM's CR-0009; (3) NEC's CR-033; (4) LGE's CR-0001; (5) SAM.AO's CR-0010
2024.11.26	05.00.02	Editorial changes reflecting comments received during WG3 review process
2024.11.27	05.00.03	Editorial changes reflecting a comment received during WG3 review process
2024.12.06	06.00	New feature: Add DL and UL Transmitted Data Volume for MAC SDU
2025.07.08	06.00.01	Adopt (1) LGE's CR-0004; (2) LGE's CR-0005
2025.07.09	06.00.02	Editorial changes reflecting a comment received during WG3 review process
2025.07.11	06.00.03	Editorial changes reflecting a comment received during WG3 review process
2025.07.28	07.00	New feature: Add a new event style for supporting Measurement-based "Event" reporting

